Ganoderma lucidum and its pharmaceutically active compounds

Bojana Boh[1], Marin Berovic[2,*], Jingsong Zhang[3] and Lin Zhi-Bin[4]
[1]*Faculty of Natural Sciences and Engineering, University of Ljubljana, Vegova 4, 1000 Ljubljana, Slovenia*
[2]*Department of Chemical and Biochemical Engineering, Faculty of Chemistry and Chemical Technology, University of Ljubljana, Askerceva 5, 1001 Ljubljana, Slovenia*
[3]*Institute of Edible Fungi, Shanghai Academy of Agriculture Sciences, Shanghai, P.R. China*
[4]*Department of Pharmacology, Peking University Health Science Center, Beijing 10083, P.R.China*

Abstract. *Ganoderma lucidum* is a wood-degrading basidiomycete with numerous pharmacological effects. Since the mushroom is very rare in nature, artificial cultivation of fruiting bodies has been known on wood logs and on sawdust in plastic bags or bottles. Biotechnological cultivation of *G. lucidum* mycelia in bioreactors has also been established, both on solid substrates and in liquid media by submerged cultivation of fungal biomass. The most important pharmacologically active constituents of *G. lucidum* are triterpenoids and polysaccharides. Triterpenoids have been reported to posses hepatoprotective, anti-hypertensive, hypocholesterolemic and anti-histaminic effects, anti-tumor and anti-engiogenic activity, effects on platelet aggregation and complement inhibition. Polysaccharides, especially β-D-glucans, have been known to possess anti-tumor effects through immunomodulation and anti-angiogenesis. In addition, polysaccharides have a protective effect against free radicals and reduce cell damage caused by mutagens.

Keywords: *Ganoderma lucidum*, cultivation, wood logs, sawdust, solid-state cultivation, submerged cultivation, triterpenoids, polysaccharides, β-D-glucans, pharmacological effects, anti-cancer effects, immunomodulation.

Introduction

Ganoderma is a white rot wood-degrading basidiomycete with hard fruiting bodies. *G. lucidum* (W.Curt.:Fr.) Lloyd and *Ganoderma applanatum* (Pers.) Pat. (Aphyllophoromycetideae) are two species most often reported as a source of medicinal compounds. In Asian traditional medicine, the fruiting body of *G. lucidum* (Fig. 1), called *Ling-Zhi* in Chinese and *Reishi* in Japanese language, has been used for treatment of several diseases for thousands of years, as reported in Shen Nong's Materia Medica [1,2]. However, an increasing systematic research (Fig. 2) into the *Ganoderma* active compounds elucidates its numerous pharmacological effects, such as antitumor, immunomodulatory, cardiovascular, respiratory, antihepatotoxic and central nervous system effects. Modern uses of *Ganoderma* therefore include treatment of coronary heart diseases, arteriosclerosis, hepatitis, arthritis, nephritis, bronchitis, asthma, hypertension, cancer and gastric ulcer [1,3]. Publications also report on *Ganoderma* antiallergenic constituents [4], immunomodulatory

*Corresponding author: Tel: 386-1-2419510. Fax: 386-1-4760-300.
E-mail: marin.berovic@fkkt.uni-lj.si (M. Berovic).

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Fig. 1. Fruiting body of *Ganoderma lucidum* (MZKI G97) originally isolated from the Slovenian forest.

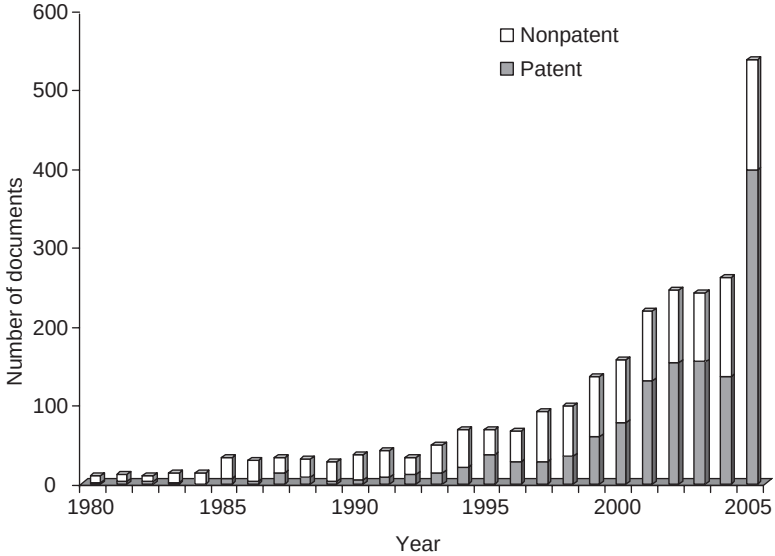

Fig. 2. Number of new documents on *Ganoderma* in the Chemical Abstracts Plus database. An increasing growth of publications indicates an accelerated research in *Ganoderma* mushrooms. The larger proportion of patents vs. non-patent publications during the last years suggests that a shift happened from basic to more applied research and development of new cultivation technologies, pharmaceutical products and nutriceutical formulations.

action [5,6], antitumor activity [7], cardiovascular effects [8], liver protection and detoxification, and effects on nervous system [9]. New reports emphasize its potential in treatment of viral, especially HIV infections [10–15].

G. lucidum is very scarce in nature. As the demands of international markets for *G. lucidum* fruiting bodies and/or mycelium biomass are in constant increase, an artificial cultivation has become essential. Successful farming on wood logs and in bags filled with wood or straw substrates has been known for decades, especially in China. Biotechnological cultivation in bioreactors on solid substrates, or with submerged liquid substrate cultivation has been developed and introduced for small and pilot-plant production [16–20]. The quality and content of physiologically active substances vary from strain to strain, and also depend on location, culture conditions [21], the growth stage of the fungus [22], the processing procedures, and formulation preparation [23].

Diverse groups of chemical compounds with pharmacological activity have been isolated from the mycelium and fruiting body of *Ganoderma* species: triterpenoids, polysaccharides, proteins, amino acids, nucleosides, alkaloids, steroids, lactones, fatty acids and enzymes [1,3]. The most important pharmacologically active constituents of *Ganoderma* mushrooms are triterpenoids and polysaccharides.

G. lucidum cultivation methods

As *G. lucidum* is very scarce in nature, artificial cultivation has become essential to meet the demands of international markets (Fig. 3).

The main traditional *G. lucidum* fruiting body cultivation methods remain sawdust cultivation in bags or bottles and cultivation on natural logs. Both cultivation technologies depend on the same essential environmental factors, including temperature, humidity and oxygen [24]. During the spawn run, mycelia grow at 10–38°C, with the optimum mycelial incubation temperature between 25°C and 32°C. The optimum moisture content of sawdust substrate is 65–70% and that of log is around 40%. pH 4.2–5.3 is regarded as optimum. The mycelial growth does not necessarily need light. Oxygen is indispensable to mycelial growth since *G. lucidum* is a strict aerobe. In the next cultivation stage of primordia formation, *G. lucidum* fruits and develops at 20–34°C, with the optimum temperature 27–32°C. The humidity of the growing room should be maintained at about 90% during primordial induction, 70–80% during cap formation and 30–40% during the final stage of fruit body development. Light (50–450 lux) is required during primordial formation and fruit body development. After the cap is formed, the growing room has to be well ventilated.

Chen [25] published detailed information on substrate formulation for *G. lucidum.* As *G. lucidum* is a lignin-degrading white-rot fungus on hardwoods, woody tissue, such as sawdust, is a natural substrate. Thiamin

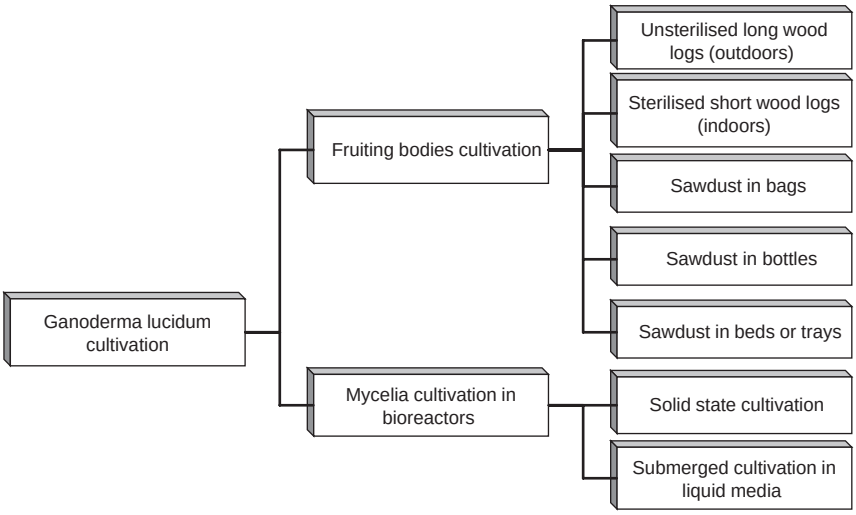

Fig. 3. Main cultivation methods for the production of *G. lucidum* fruit bodies and mycelia.

contained in fresh, unprocessed coarse bran is required for mushroom formation. A low content of sugar (1% sucrose) triggers formation and activation of lignin-decomposing enzymes. Calcium appears to encourage mushroom differentiation. Water-logging in substrate prevents air exchange and cuts off oxygen supply. If the sawdust particle size is too fine, proper air exchange is impeded. On the other hand, rough wood chips in substrate may puncture the bag and invite contamination.

Cultivation of fruit bodies on natural wood logs

Cultivation on long unsterilized logs

In the past, natural logs as long as one meter were used without sterilization in growing *Ganoderma* species in China. Fruiting body cultivation on long wood logs took much labor. Long incubation periods (2–3 years) were required to obtain mature fruiting bodies on such substrates [24,26].

Cultivation on short sterilized logs

Since the late 1980s, new trends have been developed using short logs. Almost all *Ganoderma* spp. natural log growers adopted the short-log cultivation in China, Japan, the United States and elsewhere. High yield in a shorter cultivation time enabled quicker turnover of the capital. Short-log cultivation

takes only 4–5 months for mycelial incubation, and the fruiting body can be cropped in the same year. Detailed procedures on short-log cultivation were published elswere [24,26–28].

The main cultivation stages of growing *G. lucidum* on short natural logs enclosed in ventilated synthetic bags during spawn run include [26]:

- *Preparation of wood logs*. Most broad-leaf hardwoods can be used, *t*he standard log size is 15 cm in diameter, and 15–24 cm long. Moisture content is 35–40%.
- *Enclosing logs into bags, sterilization*. Heat-sealed polypropylene or polyethylene bags with micro-filter windows are used.
- *Spawning*. A variety of spawns, such as pure culture liquid mycelial spawn, grain spawn and sawdust-bran spawn can be used. Usually 5–10 g of spawn is used for each log.
- *Spawn run* is carried out in darkness, and less oxygen is required. Special attention should be given to ensure proper mycelial colonization in the log. Lack of oxygen or poor aeration, such as water-logging, results in poor mycelial growth and slow growth rate.
- *Primordia initiation. Ganoderma* spp. primordia are usually formed 50–60 days after spawning. Brief exposure to very little light triggers *Ganoderma* spp. primordia initiation. Oxygen is also conducive to primordia formation.
- *Embedding in soil*. Colonized logs are embedded vertically directly in soil after primordia formation, leaving the primordia above the ground level. To retain moisture, the soil is covered with chopped straw.
- *Maintaining suitable growth parameters*. The most crucial factor during primordia initiation is to have high humidity, preferably 90–95%, while the most crucial factor during pileus differentiation in fruiting, is increase in ventilation to reduce CO_2 build-up from the drastic increase in respiration from *Ganoderma* spp. fruiting. Differentiation of *Ganoderma* spp. fruiting is highly sensitive to CO_2 concentration, which determines whether antler-shaped fruiting bodies ($CO_2 > 0.1\%$), or fruiting bodies with a well-formed pileus ($CO_2 < 0.1\%$) will be produced. CO_2 concentration at 0.04–0.05%, as close to fresh air as possible (0.03% CO_2), should be maintained for production of pileated mushrooms (mushrooms with caps). Air humidity can be supplied by a fine mist (1–2, or 3–4 times/day).
- *Harvesting mushrooms*. From primordia formation to fruiting body for harvest takes approximately 25 days. Fruiting maturity is indicated by the disappearance of the undifferentiated white growth at the edge of the fruiting body. Cultivation is then continued at reduced air humidity of 60–85% for additional 7–10 days to encourage further growth in pileate thickness and firmness. Harvest is done by cutting the stipe (stalk); only 2 cm of the stipe remains with the pileus.

- *Post-harvest treatment* includes immediate drying under the sun or with heat (60°C) for 2–3 days. Improper drying lowers the quality of the product.

Cultivation of fruit bodies on sawdust substrates

Sawdust substrates in sterilizable bags (synthetic log cultivation)

According to Royse [29], most cultivation of *G. lucidum* is on supplemented sawdust contained in heat-resistant polypropylene bottles or bags. Sawdust of hardwoods is usually supplemented with rice bran (10%) and $CaCO_3$ (3%), moistened with water and filled (700 g) into plastic bags. A plastic collar then is fitted onto each bag and closed with a cotton plug. After heat treatment (95–100°C for 5 h) the substrate is allowed to cool overnight and then inoculated with grain or sawdust spawn. The inoculated substrate is incubated for 3–4 weeks or until the spawn has fully colonized the substrate. Mushroom production is initiated by maintaining air temperature at about 28°C with relative humidity in the range 85–90%. Basidiocarps begin to appear in about 1–2 weeks after initiation. Approximately 2–3 months after the appearance of primordia, mushrooms are ready to harvest. A mushroom is considered mature when the whitish margin around the edge of the basidiocarp has turned red. The substrate may yield another harvest of mushrooms after removal of the first flush.

Chen [25] recommended the following substrate formulation for *Ganoderma* cultivation: oak sawdust 80%, fresh unprocessed coarse wheat bran 18%, supplemented with sucrose 1%, calcium carbonate (or calcium sulphate) 1%, with approximately 67–70% water. For 500 g dry substrate (containing 400 g oak sawdust and 90 g wheat bran), 1 l of water containing 5 g of sucrose and 5 g of calcium carbonate was added. Using this formulation, scale-up cultivation by mushroom growers in the United States and Canada has been successful. Another extensive article on *Ganoderma* cultivation in bags (synthetic logs) is available in [30].

Several publications describe *G. lucidum* cultivation in bags under unconventional conditions. A Japanese patent [31] claimed cultivation of *G. lucidum* fruit bodies in an antler form in bags. Nascimento [32] grew *G. lucidum* on hardwood chips and sawdust of two Chilean native red wood trees, *Nothofagus obliqua* and *Nothofagus alpine*, in polypropylene bags. No difference was observed in the fruiting stage between the two species of tree wood. Gonzalez-Matute *et al.* [33] studied the sunflower seed hulls as a main nutrient source for the cultivation of *G. lucidum* in bags (synthetic log system). The study concluded that sunflower seed hulls could be used as the main energy and nutritional source in substrates for *G. lucidum* cultivation, and the addition of 5% malt to the substrate improved the mushroom growth rate.

Yang *et al.* [34] utilized stillage grain from a rice-spirit distillery in the cultivation of *G. lucidum* in polypropylene bags.

Due to its high content of carbohydrate and nitrogen, stillage grain was considered as a nutritive substrate for mycelial growth. Sawdust supplemented with stillage grain at a ratio 4:1 at a water content of 60% was optimal for the production of fruiting body. Hsieh *et al.* [35] used the soy residue from the waste of tofu manufacturing to culture *G. lucidum* in polypropylene plastic bags. The fruiting bodies were fully developed only for the C/N ratios of 70 and 80.

Sawdust substrates in bottles and pots

Kim [36] grew twenty-one isolates of nine *Ganoderma* species (including *G. lucidum*) in 21 sterilizable plastic bottles on solid substrates based on sawdust. The substrate was prepared by mixing oak sawdust and wheat bran (8:2, v/v), and the water was added to 65% of the total volume. The cultivation room was maintained at 28–31°C and 85% of relative humidity until primordial formation and ventilated once or twice for 10–20 min afterwards. After the pileus was formed, ventilations were frequent (5–6 times/day) and the relative humidity was controlled at 80–85%.

A Japanese patent [37] claimed an arrangement for cultivating *G. lucidum* or other mushrooms in bottle containers, where a negative voltage was applied to the container body for activation and production of mushrooms of excellent quality in a short period. Another Japanese patent described a cultivation method for *Ganoderma* in pots by protecting a nursery bed against the infiltration of bacteria by a plastic cover. In a work by Shigeru [39], finely crushed thinned citrus fruits and residue of citrus fruit juice were utilized in the culture medium, mixed with rice bran at a weight ratio of about 10:2 for *G. lucidum* in bottles. *Ganoderma* was cultured at about 25–30°C and about 60–90% humidity to develop fruit bodies.

Sawdust substrates in trays or beds

Chen [28] reported on *G. lucidum* cultivation on trays or beds in North America. The article stated that wood-chip or sawdust beds were labor saving, if contamination was avoided. The substrate, 12 cm deep, composed of wood chips supplemented with sawdust, was spread evenly over cultivation trays or beds. Colonized sawdust, grain, or liquid spawn, 0.5 cm thick or more, was seeded on the surface under still air and covered with a plastic sheet. After 3–5 days, a layer of white mycelia became visible and begun to penetrate the substrate. When primordia were formed after 1–2 months, the plastic cover sheet was removed. The growing space was maintained at one-third diffuse light, 85–95% relative humidity and 25°C. Air circulation or aeration was provided 3–5 times/day 5–10 min each time.

Cultivation of *G. lucidum* mycelia in a bioreactor

Solid-state cultivation

A Slovenian patent [40] claimed a process of growing *G. lucidum* on a solid cultivation substrate using the solid-state cultivation in a horizontal stirred bioreactor. Beech sawdust was used as a solid cultivation substrate. The process enabled a precise leading and monitoring of the fungal growth at sterile conditions. Large quantities of biomass could be prepared according to the process to yield products applicable to pharmacy. The biomass could also be used as a solid inoculum for the further cultivation of *G. lucidum.*

A World patent application with a Chinese priority [41] described a method for propagating fungi and producing fungal metabolites with medicinal activities, using a solid-state fermentation process and cultivation in bottles. The invention also described substrates for small- and large-scale fungal cultivation of *G. lucidum*, *Cordyceps sinensis*, *Antrodia camphorata*, *Trametes versicolor and Agaricus blazei*.

Chen [28] reported that in North America, *Ganoderma* mycelial preparations for human consumption are produced by solid-state fermentation on grain- or soy-based substrates.

Submerged cultivation in liquid media

Authors use different substrate compositions for the submerged cultivation of *G. lucidum* mycelia. An example given in [28] consisted of sucrose 50.0 g, ammonium succinate 3.2 g, KH_2PO_4 1.0 g, $MgSO_4 \cdot 7H_2O$ 0.3 g, $FeSO_4 \cdot 7H_2O$ 13.0 mg, $ZnSO_4 \cdot 7H_2O$ 4.0 mg, yeast or malt extract 10.0 g, adjusted to pH 5.2 with concentrated ammonia, water to 1 l.

An optimized medium for a shake-flask submerged culture of *G. lucidum*, reported by Chang *et al.* [42] consisted of 1.88 g/l $CaCO_3$, 71.4 brown sugar, 12.1 g/l malt extract, 2.28 g/l yeast extract, 18.4 g/l skim milk, 3.44 g/l safflower seed oil, 3.96 g/l olive oil, at pH 6.5. Compared to an unoptimized substrate, the mycelium formation was markedly improved from 1.70 g/l to 18.70 g/l, and the polysaccharide production increased from 0.140 g/l to 0.420 g/l.

Hsieh *et al.* [43] studied production of polysaccharides by *G. lucidum* in shake flasks under various limitations of nutrients, including carbon-source, nitrogen-source, phosphate-source, magnesium-source and dissolved oxygen. Different responses of polysaccharide production were observed under different limitations of nutrients.

A Slovenian patent [44] described a procedure for *G. lucidum* inoculum preparation in a shaked culture, and the production of mycelia by submged fermentation in a bioreactor. According to the patent description, potato dextrose agar of the total area of 100–200 mm^2 overgrown by the fungus

G. lucidum was transferred into a 500 ml Erlenmayer flask containing 100 ml of substrate. The vegetative substrate contained the filtrate of 300 g/l of peeled cooked potato, 20 g/l of glucose and 2% v/v of olive oil, and was filled with distilled water of pH 5.8 to the total volume of 1 l. The culture was shaken in this medium for 80–160 h at the temperature of 20–30°C at 80–160 rotations per minute. In a bioreactor, the substrate was sterilized at the temperature of 110–130°C and with mixing 200–400 rotations per minute, for about half an hour. After cooling down to 30°C, the sterile substrate in a bioreactor was inoculated with 17% v/v of the vegetative inoculum containing mycelium of *G. lucidum*, produced in a shaken culture of the age of 120–170 h. The growth of the mycelium lasted for 160 h during which the concentration of dissolved oxygen was maintained by aeration of 6–15 l/min at 200–600 rotations/min, maximal redox potential during growing amounting to 410–460 mV and minimal partial oxygen pressure from 25 to 33% v/v at pH from 4.10 to 4.30.

Yang and Liau [45] studied the influence of cultivation parameters on polysaccharide formation by *G. lucidum* in submerged cultures. The substrate consisted of glucose 50 g/l; K_2HPO_4 0.5 g/l; KH_2PO_4 0.5 g/l; $MgSO_4 \cdot 7H_20$ 0.5 g/l; yeast extract 1 g/l; and ammonium chloride 4 g/l. Optimal temperature was 30–35°C and the pH 4–4.5. Polysaccharide concentration reached 1.6 mg/ml. Agitation and aeration influenced the formation and secretion of polysaccharides. The optimal rotating speed was 150 rpm in 7-day flask cultures, while the agitation speed employed in fermenter culture greatly affected the production rate and maximum concentration of polysaccharide. Although higher speeds enhanced mixing efficiency and polysaccharide release, higher shear stress had a detrimental effect on mycelial growth and polysaccharide formation.

Lee [46] reported that pH control substantially affected mycelial cell growth and exopolysaccharide production of the mycelial cultivation of *G. lucidum*. The fermentor was a concentric draught-tube internal loop type which can greatly reduce shear stress as compared to an impeller-type fermentor. Five percent (v/v) of the culture was inoculated into the fermentor and cultivated at 25°C with air being supplied at a rate of 2.5 vvm under batch conditions. Compared to the uncontrolled pH cultivation, the bistage pH control technique, in which pH was shifted from 3 to 6 at the initial phase of the exponential growth, increased exopolysaccharide production from 4.1 g/l to 20.1 g/l, retained the desirable morphologies of the mycelia during cultivation, and resulted in low viscosity and yield stress of the culture broth.

Fang and Zhong [47] investigated the effects of initial pH on simultaneous production of ganoderic acid and a polysaccharide by *G. lucidum*. An initial pH value, varying within the range 3.5–7.0, had a significant effect on the cell growth and product biosynthesis. At an initial pH of 6.5, a maximum in biomass of 17.3 ± 0.12 g/l by dry weight was obtained, as well as a maximal

specific production of ganoderic acid of 1.20 ± 0.03 mg/100 mg dry weight and total production of 207.9 ± 2.7 mg/l. Lowering the initial pH from 6.5 to 3.5 gradually led to a higher production of extracellular polysaccharide and a higher specific production of intracellular polysaccharide. The same research group [48] studied the effects of nitrogen source and initial glucose concentration in submerged fermentation of *G. lucidum* for simultaneous production of bioactive ganoderic acids and polysaccharides. The cells could not grow well when either yeast extract or peptone was used as the sole nitrogen source. However, a combined addition of 5 g/l of yeast extract and 5 g/l of peptone was optimal for the cell growth and metabolite production. Initial glucose concentration within 20–65 g/l greatly affected the cell growth and product biosynthesis. The highest levels of cell density (16.7 g dry weight/l), intracellular polysaccharide (1.19 g/l) and ganoderic acid (212.3 mg/l) were obtained at an initial glucose concentration of 50 g/l.

Fang *et al.* [49] also reported on a significance of inoculation density control in production of polysaccharide and ganoderic acid by submerged culture of *G. lucidum.* Control of inoculation density was significant for cell growth, morphology, and production of polysaccharide and ganoderic acid. A maximal cell concentration of 15.7 g dry cell weight/l was obtained at an inoculation density of 330 mg dry weight/l. For inoculation density within the range 70–670 mg dry weight/l, a large inoculation density led to a small pellet size and high production of extracellular and intracellular polysaccharides, while a relatively big pellet size and high accumulation of ganoderic acid were observed at a low inoculation density. It was also shown that small pellet size resulted in high polysaccharide production, while large pellet size led to high production of ganoderic acid.

In a work of Berovic *et al.* [20] *G. lucidum* was cultivated in a liquid substrate based on potato dextrose and olive oil. The influences of inoculum and oxygen partial pressure in batch and fed-batch cultivation in a 10 l laboratory-stirred tank reactor were studied. The cultivation conditions were temperature of cultivation, $T = 30°C$; mixing, $N = 300\,min^{-1}$; aeration, $Qg = 10\,l\,min^{-1}$; average values of pH, 5.8–4.2; oxygen partial pressure, 70–80% and redox potential, Eh = 300–400 mV. Fungal biomass was found to be oxygen and shear sensible. Using a 17% (wet weight) 6 days old vegetative inoculum, $9.6\,g\,l^{-1}$ of dry biomass in batch cultivation and $15.2\,g\,l^{-1}$ in fed-batch process were obtained. Extracellular ($9.6\,g\,l^{-1}$) and intracellular ($6.3\,g\,l^{-1}$) polysaccharide fractions were isolated. Extracellular polysaccharide fraction and four intracellular polysaccharide fractions were obtained. Polysaccharides were further separated by ion-exchange, gel and affinity chromatography. The isolated polysaccharides were mainly β-D-glucans. Immunostimulatory effects of isolates were tested on induction of cytokine (tumor necrosis factor a (TNF-a) and interferon g (IFN-g)) synthesis in primary cultures of human peripheral blood mononuclear cells (PBMC) isolated from a buffy coat. The TNF-a-inducing activity was comparable with

romurtide, which has been used as a supporting therapy in cancer patients treated with radiotherapy and/or chemotherapy.

Tang and Zhong [50] studied the effects of carbon source and initial sugar concentration on the production of ganoderic acids and polysaccharided in a fed-batch *G. lucidum* cultivation process in shake flasks and in a stirred bioreactor. Sucrose as a carbon source was suitable for the extracellular polysaccharide production although the cells could not grow well. Lactose was beneficial for the cell growth and production of ganoderic acid and intracellular polysaccharides. When the initial lactose concentration exceeded 35 g/l, the ganoderic acid accumulation was decreased. The ganoderic acid production was remarkably improved by pulse feeding of lactose, when its residual concentration was between 10 g/l and 5 g/l.

Submerged fermentation of *G. lucidum* is viewed as a fast and cost-effective alternative for efficient production of polysaccharides and ganoderic acids from *G. lucidum*. However, submerged cultivation of mycelia is characterized by an increase in broth viscosity with time, which is a consequence of increased cell concentration and the accumulation of extracellular polysaccharides that dramatically alter the rheological characteristics of fermentation broth, and create a series of problems that have to be solved, especially oxygen supply. Oxygen affects cell growth, cellular morphology, nutrients uptake and metabolite biosynthesis. Tang *et al.* [51] reported on the effects of oxygen supply on the submerged cultivation of *G. lucidumin* in a 3.5 l agitated bioreactor with two six-bladed turbine impellers. Aeration was through a ring sparger with a pore size of 0.8 mm. Fermentation was conducted at 30°C in the dark. The cultivation medium consisted of 35 g/l lactose, 5 g/l peptone, 2.5 g/l yeast extract, 1 g/l $KH_2PO_4 \cdot H_2O$, 0.5 g/l $MgSO_4 \cdot 7H_2O$ and 0.05 g/l Vitamin B1. The results showed that an initial volumetric oxygen transfer coefficient (K_La) value within the range 16.4–96.0 h^{-1} had a significant effect on the cell growth, cellular morphology and metabolites biosynthesis. An increase of initial K_La led to a bigger size of mycelia aggregates and a higher production of ganoderic acids.

Fang *et al.* [49] studied the significance of inoculation density and pellet size in the submerged culture of *G. lucidum* on production of polysaccharides and ganoderic acids. Inoculum sizes of 70, 170, 330 and 670 mg dry cell weight (DW)/l were tested. Inoculation density significantly affected the cultivation process. Small pellet size resulted in high polysaccharide production, while large pellet size led to high production of ganoderic acid. Pellets with diameters smaller than 1.2 mm, 1.2–1.6 mm and larger than 1.6 mm had ganoderic acid content of 0.98, 1.27 and 1.62 mg/100 mg DW, respectively.

Some authors reported on submerged cultivation of *G. lucidum* mycelia on unconventional substrates, including liquid waste materials, such as thin stillage and deproteinated cheese whey.

Hsieh *et al.* [52] produced *G. lucidum* polysaccharides by reusing thin stillage (from a wine manufacturing plant) in a shake flask culture. By

adjusting the pH 5, a 60% thin stillage was used successfully to grow the mycelia of *G. lucidum* with the highest cell concentration of 7.8 g/l and polysaccharide production of 7.50 g/l. Molasses addition produced the highest mycelia growth rate and a cell concentration increase to 12.7 g/l. Glucose addition led to increase overall polysaccharides production up to 3.69 g/l. Polysaccharides in the range of molecular weight from 10,000 to 200,000 Da were also found at almost three times production with thin stillage only.

Lee *et al.* [53,54] utilized deproteinated cheese whey for cultivating mycelia of the *G. lucidum* in a bioreactor by submerged cultivation and concluded that cultivation of *G. lucidum* mycelia could offer a potential cost-effective solution for an alternative utilization of the deproteinated cheese whey.

Main pharmacologically active compounds in *G. lucidum*

The most important pharmacologically active constituents of *Ganoderma* mushrooms are triterpenoids and polysaccharides (Fig. 4).

Triterpenoids from *Ganoderma* mushrooms

Over 150 triterpenoids were found in *Ganoderma* spp., such as ganoderic (highly oxygenated C_{30} lanostane-type triterpenoids), lucidenic, ganodermic, ganoderenic, ganolucidic and applanoxidic acids, lucidones, ganoderals and ganoderols [5–64]. Representative examples are shown in Figs. 5–13.

Boh *et al.* [65] reported that the quantity of triterpenoids differed in older and younger parts of *G. applanatum* fruiting bodies. The highest amount of triterpenoid acids was found in the tubes (6.4 mg/g of air-dried weight), followed by the younger dark context layer of the pileus (2.5 mg/g), the older white context layer (0.6 mg/g) and the upper surface of the fruiting body (0.6 mg/g).

Triterpenoids have numerous pharmacological effects, as summarized below:

Anti-hepatotoxic and hepatoprotective effects

Hirotani *et al.* [55] successfully isolated ganoderic acids R and S from the cultured mycelia and proved their strong anti-hepatotoxic activity in the galactosamine-induced cytotoxic test with primary cultured rat hepatocytes. Kim *et al.* [66] reported on beta-glucuronidase-inhibitory and hepatoprotective effect of *G. lucidum*.

Anti-tumor effects

Isolated ganoderic acids Z, Y, X, W, V and T from *Ganoderma* mycelia demonstrated cyctotoxic activitiy *in vitro* on hepatoma cells [67].

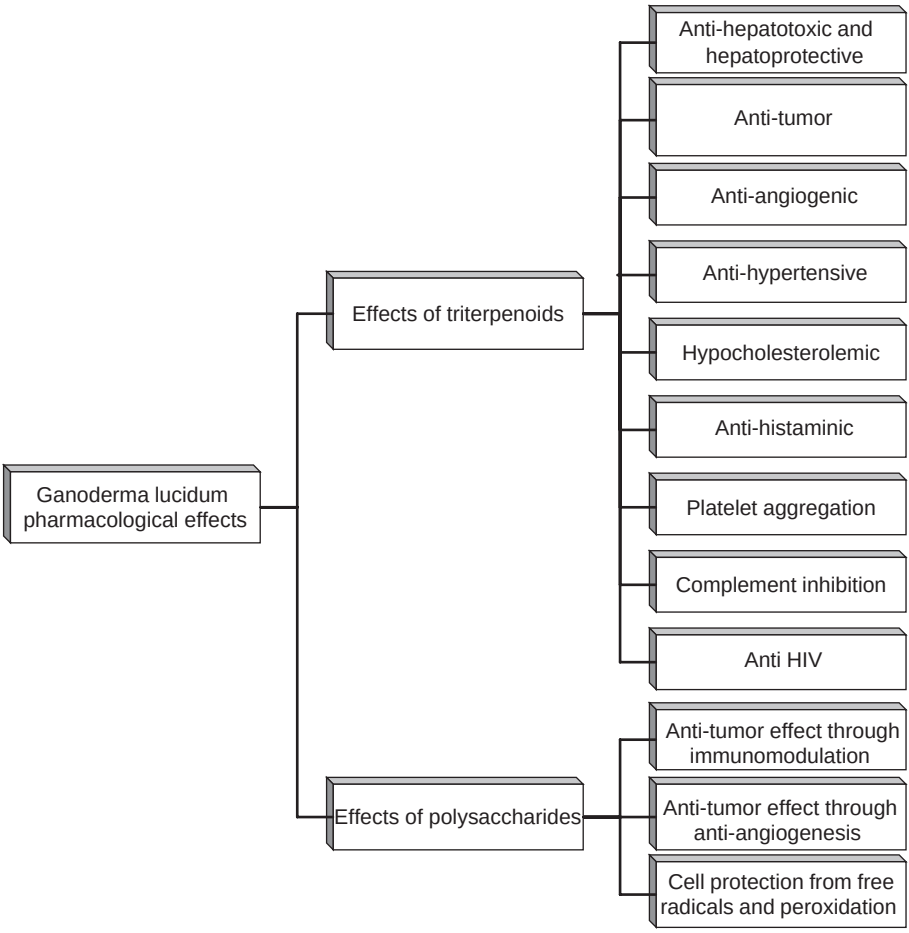

Fig. 4. Main pharmacological effects of *G. lucidum*.

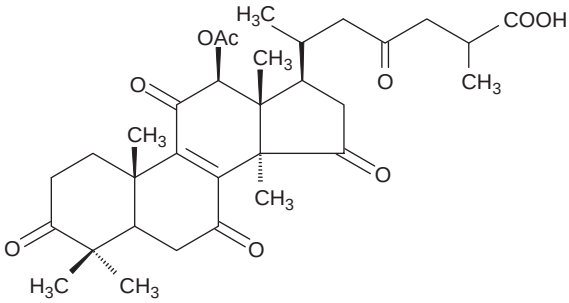

Fig. 5. Ganodermic acid F (12β-acetoxy-3,7,11,15,23-pentaoxo-5α-lanost-8-en-26-oic acid).

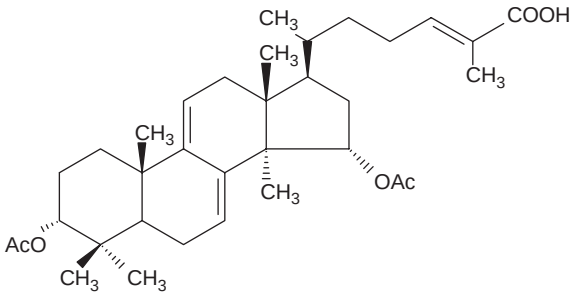

Fig. 6. Ganodermic acid R ((24E)-3a,15a-diacetoxy-5a-lanosta-7,9(11),24-triene-26-oic acid).

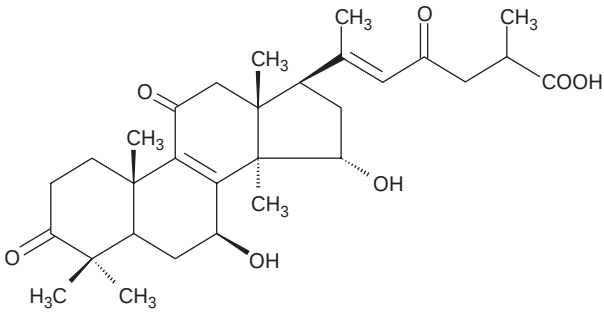

Fig. 7. Ganoderenic acid A ((20E)-7β,15α-dihydroxy-3,11,23-trioxo-5α-lanosta-8,20-dien-26-oic acid).

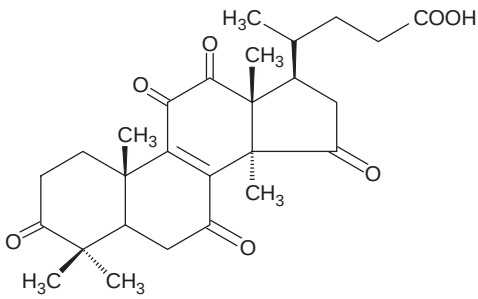

Fig. 8. Lucidenic acid D1 (4,4,14α-trimethyl-3,7,11,12,15-pentaoxo-5α-chol-8-en-24-oic acid).

Lin *et al.* [68] reported that a triterpene fraction from *G. lucidum* significantly inhibited growth of human hepatoma Huh7 cells, probably due to the oxidative stress induction. The same triterpenoid extract only had little influence on normal human liver cell line.

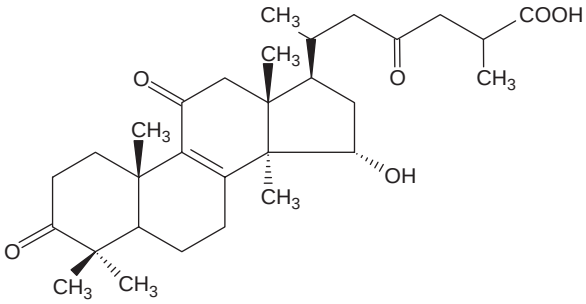

Fig. 9. Ganolucidic acid A (15a-hydroxy-3,11,23-trioxo-5a-lanost-8-en-26-oic acid).

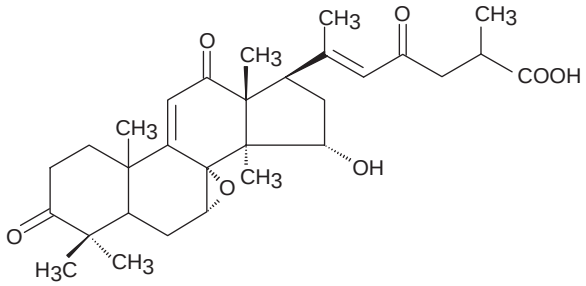

Fig. 10. Applanoxidic acid A ((20E)-15α-hydroxy-7α,8α-epoxy-3,11,23-trioxo-5α-lanosta-9(11),20-dien-26-oic acid).

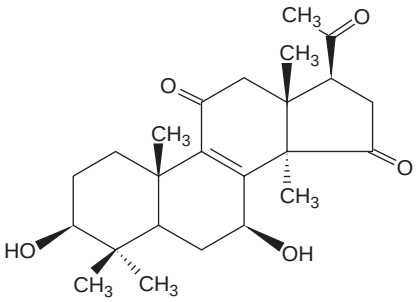

Fig. 11. Lucidone (3b,7b-dihydroxy-4,4,14a-trimethyl-11,15,20-trioxo-5a-pregn-8-ene).

Gao *et al.* [69] isolated three new lanostante-type triterpene aldehydes, named lucialdehydes A-C, from the fruiting bodies of *G. lucidum*. Lucialdehydes B, C showed cytotoxic effects on Lewis lung carcinoma (LLC), T-47D, Sarcoma 180, and Meth-A tumor cell lines. Lucialdehyde C exhibited the most potent cytotoxicity against the tested cell lines with ED50 values of 10.7, 4.7, 7.1 and 3.8 mg/l, respectively.

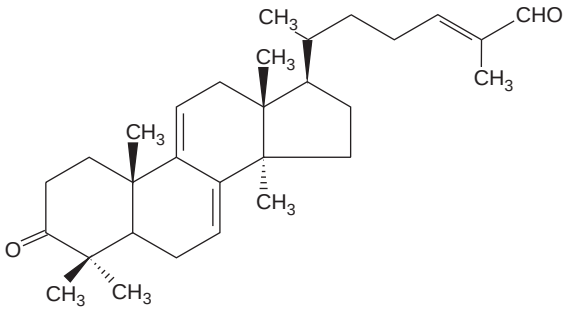

Fig. 12. Ganoderal A ((24E)-3-oxo-5a-lanosta-7,9(11),24-triene-26-al).

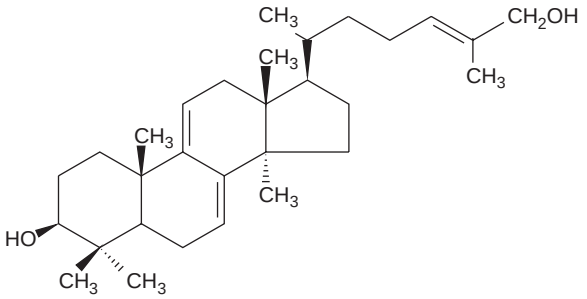

Fig. 13. Ganoderol B (ganodermadiol-5a-lanosta-7,9(11),24-triene-3b,26-diol).

Six new highly oxygenated lanostane-type triterpenes isolated from *Ganoderma* spores also showed direct cytotoxicity *in vitro* on the Meth-A and LLC tumor cell lines [70]. It has also been suggested that the triterpene-enriched fraction, WEES-G6, prepared from mycelia of *G. lucidum* inhibited the growth of human hepatoma Huh-7 cells. Treatment with WEES-G6 caused a rapid decrease in the activity of cell growth regulative protein, PKC, and the activation of JNK and p38 MAP kinases, which resulted in a prolonged G2 cell cycle phase and strong growth inhibition of the hepatoma cells [71].

The alcohol extract of *G. lucidum* also showed that it inhibited cell proliferation in a dose- and time-dependent manner, which might be mediated through up-regulation of p21/Waf1 and down-regulation of cyclin D1. Furthermore, it can directly induce apoptosis in MCF-7 cells, which might be mediated through up-regulation of a pro-apoptotic Bax protein and not by the immune system (Hu *et al.*, 1999a). Two alcohol extracts (I and III) from *G. lucidum* spores strongly inhibited the growth of HeLa cells. Moreover, extract III was shown to be capable of blocking the cell cycle at the transition from G1 to S phase and inducing a marked decrease of intracellular calcium level. These results imply that the effective extract might influence the cell

cycle and cellular signal transduction by altering the calcium transport system [72].

Yu *et al.* [73] reported on the correlation between intracellular triterpenes from mycelia of *G. lucidum* in different growth stages of submerged cultivation and the inhibition effect on K562 tumor cells. The results showed that intracellular triterpenes were produced mainly in the later period of fermentation. Intracellular triterpenes from different stages of fermentation varied in types, quantity and relative proportion, and exhibited different effects on tumor cell inhibition. Optimum culture conditions for producing intracellular triterpenes inhibitory to K562 cells were shake flask cultures.

The results of Liu *et al.* [74] indicated that the triterpenoids fraction of *G. lucidum* might be a useful ingredient in the treatment of benign prostatic hyperplasia. The ethanol extract of *G. lucidum* showed inhibitory activity on both isozymes (types 1 and 2) of 5α-reductase and suppression effects of ventral prostate growth induced by testosterone in castrated rat, but not induced by dihydrotestosterone. Activity-guided fractionation and TLC analysis suggested that the active principles *in vivo* were triterpenoids.

Mueller *et al.* [75] results indicate that *G. lucidum* extract has a profound activity against leukemia, lymphoma and multiple myeloma cells and may be a novel adjunctive therapy for the treatment of hematologic malignancies. In their work, *G. lucidum* extract was screened for its anti-proliferative activity using a panel of 26 human cancer cell lines.

Anti-angiogenic effect

Kimura *et al.* [76] reported that anti-tumor and anti-metastatic activities of a triterpenoid fraction of *G. lucidum*, containing ganoderic acid F, were due to the inhibition of tumor-induced angiogenesis. The triterpenoid fraction (100 and 200 mg/kg) of the fruit bodies of *G. lucidum* inhibited primary solid-tumor growth in the spleen, liver metastasis and secondary metastatic tumor growth in the liver in intrasplenic Lewis lung carcinoma (LLC)-implanted mice. In addition, the triterpenoid fraction (800 mμg/ml) inhibited angiogenesis induced by Matrigel (a soluble basement membrane extract of the Engelbreth-Holm-Swarm tumor) supplemented with vascular endothelial growth factor (VEGF) and heparin in an *in vivo* model.

Anti-hypertensive effects

Morigiwa *et al.* [57] found out that some *G. lucidum* triterpenes inhibited angiotensin converting enzyme, while Kabir *et al.* [77] reported on dietary effects of *G. lucidum* on blood pressure and lipid levels in spontaneously hypertensive rats.

Hypocholesterolemic effects

Lin and Shiao [60] reported on inhibitory effects of ganoderic acid Mf and ganodermic acid T-O on cholesterol synthesis.

Anti-histaminic effects

Kohda *et al.* [78] studied biologically active constituents of *G. lucidum* and found out that triterpenes, such as ganodermic acids C and D, inhibit histamine release.

Platelet aggregation effects

Wang *et al.* [79,80] reported that ganodermic acid S exhibited amphipathic effect on the platelet aggregation. At high concentration of ganodermic acid S platelet aggregation occurred, while at a low concentration the aggregation was inhibited. The inhibition was concentration and time dependent.

Su *et al.* [81,82] reported that collagen-induced platelet aggregation by ganodermic acid S was due to blocking of Ca mobilization through the thromboxane A2-dependent pathway in human platelets response to collagen, and that ganodermic acid S elevated prostaglandin E(1)-induced cyclic AMP in human platelets.

Complement inhibition effect

Min *et al.* [83] reported that ganoderiol F, ganodermanondiol, ganodermanontriol from spores of *G. lucidum* exhibited strong anti-complement activity, through which these substances can affect the humoral immune system in the host defense.

Anti-HIV activity

In 1997, Kim *et al.* [12] reported that the water-soluble extract of *G. lucidum* inhibited cytopathic effect of HIV-1, and Hattori *et al.* [13] reported on inhibitory effects of components from *G. lucidum* on the growth of human immunodeficiency virus (HIV) and its protease activity. El-Mekkway *et al.* [14] studied anti-HIV-1 and anti-HIV-1 protease substances from *G. lucidum*. Ganoderiol F and ganodermanontriol were active as anti-HIV-1 agents in concentration 7.8 μg/ml. Several other *Ganoderma* triterpenoids moderately inhibited HIV-1 protease activity at concentrations of 0.17–0.23 mM. Similar results with another set of *G. lucidum* triterpenoids were reported by Min *et al.* [15]. Ganoderic acid β, ganodermanondiol, ganodermanontriol, ganolucidic acid A and lucidumol B showed strong anti-HIV-1 protease activity with IC_{50} vaules of 20–90 μM.

Polysaccharides from *Ganoderma* fruit bodies and mycelia

In recent years, a lot of scientific attention has been focussed on *Ganoderma* polysaccharides, which represent a structurally diverse class of biological macromolecules with a wide range of physicochemical properties. Studies have shown that the most active immunomodulatory polysaccharides are water-soluble β-1-3-D and β-1-6-D glucans, that can be precipitated by ethanol. Their prevailing structure is β-1-3 D-glucopyronan with 1–15 units of β-1-6 monoglucosyl side chains. Their 1,3-linked backbone, relatively small side chains and an organized helical structure are beneficial for the immunostimulation [84]. Other immunomodulatory polysaccharides have been reported, especially glycopeptides [85] and proteoglycans [86].

Reports on the pharmacological activity of *Ganoderma* polysaccharides mainly focus on anti-tumor effects, which are linked to immunomodulation, although other effects, such as regulation and protection of cells, have also been observed. Bioactive water-soluble polysaccharides have been isolated from the fruiting bodies and from the mycelial biomass cultivated in liquid culture. Few have been isolated from the culture medium. Some water insoluble anti-tumor polysaccharides were also identified [87].

Anti-tumor effects and immunological mechanisms

In 1971 Sasaki *et al.* [88] reported on anti-tumor polysaccharides from some Polyporaceae, including *G. applanatum*. The study of anti-tumor effects of *G. lucidum* and their mechanisms had become matters of great interest. Since 1980s, numerous pharmacological investigations demonstrated that the hot water extract of *G. lucidum* polysaccharides inhibited the tumor growth in several tumor-bearing mice models [89–95], but the mechanism of the anti-tumor effect of *G. lucidum* was not well understood. A hypothesis was set up that *G. lucidum* polysaccharides exert anti-tumor activity by reinforcing body's intrinsic immune-system defending functions. This hypothesis was used as a direction to guide further research to designing protocols and experiments to confirm its validity.

Further research [96–99] demonstrated that *in vivo* water-soluble extract of *G. lucidum* and *G. lucidum* polysaccharides inhibited the growth of S-180 sarcoma and Lewis lung carcinoma implanted in mice. However, when added to the cultures of S-180 or HL-60 tumor cells, neither *G. lucidum* water extract nor *G. lucidum* polysaccharides inhibited the proliferation and induced apoptosis of the tumor cells, even at the very high concentration (400 mg/l) of *G. lucidum* polysaccharides, which indicated that either *G. lucidum* or its active fraction had no direct cytotoxicity on tumor cells.

In the 1990s it became evident that *Ganoderma* polysaccharides indeed affect the immune system. A number of reports showed that polysaccharides stimulated immune functions both *in vivo* and *in vitro*, and that macrophages

were involved in this mechanism [100]. The proliferation of cancer cells was not affected by *Ganoderma* polysaccharides alone, but were significantly inhibited by the conditioned medium from polysaccharide-activated blood mononuclear cells [101–106].

Lei *et al.* [107] studied antagonistic effects of *G. lucidum* polysaccharides on the immunosuppressive response induced by cyclosporin A, hydrocortisone and anti-tumor agents, and reported on enhancement of cell-mediated immune functions and augmentation of cytokine production. Other immunomodulation effects, *i.e.*, enhancement of unspecific immune functions, were reported [108], where *Ganoderma* polysaccharides affected intracellular free calcium and oxygen free radicals in murine peritoneal macrophages.

Anti-tumor activity of cytotoxic drug such as cyclophosphamide was enhanced by oral administration of *G. lucidum* polysaccharides. *G. lucidum* polysaccharides induced HL-60 cells apoptosis by boosting macrophage activity [96,97].

G. lucidum polysaccharides (0.8, 3.2 and 12.8 mg/l) promoted not only the maturation of cultured murine bone marrow-derived dendritic cells (DC) through increasing co-expression of CD11c and I-A/I-E molecules on DC surface and enhancing IL-12 p40 production as well as its mRNA expression in DC, but also the immune response initiation induced by DC, such as cytotoxicity of specific cytotoxic T lymphocytes (CTL) induced by DC pulsed with P815 tumor antigen during the stage of antigen presentation by augmenting protein and mRNA expression of IFN-γ and granzyme B [109–110].

Shao *et al.* [111] investigated immune receptors for polysaccharides from *G. lucidum.* The study, which was designed to identify and characterize the immune receptors for polysaccharides, demonstrated that *G. lucidum* polysaccharides activated BALB/c mouse B cells and macrophages, but not T cells, *in vitro*.

Chien *et al.* [112] reported that a fucose-containing glycoprotein fraction, isolated from *G. lucidum*, increased the population and cytotoxicity of $CD56^+$ NK-cell in human umbilical cord blood mononuclear cells.

Lei and Lin [113–114] reported that *G. lucidum* promoted the mixed lymphocyte reaction (MLC), and its active polysaccharides increased the DNA synthesis of spleen cells in MLC through the enhancement of DNA polymerase induction in mice.

The anti-tumor effect of *G. lucidum* polysaccharides was mediated by cytokines released from activated T lymphocytes and macrophages [102]. *In vitro*, *G. lucidum* polysaccharides induced B lymphocyte activation and proliferation [113,115–117]. However, studies on antibody production induced by *G. lucidum* were varied in different authors' reports, in which *G. lucidum* polysaccharides having a backbone consisting of 1,4-linked α-D-glucopyranosyl residues and 1,6-linked β-D-galactopyranosyl residues with branches at *O*-6 of glucose residues and *O*-2 of galactose residues had little effect on serum IgG and complement (C3) levels after intraperitoneal

injection at a dose of 25 mg/kg for 4 days [116], and *G. lucidum* oral infusion significantly suppressed IgG1 and increased IgG2a production in mice with airway inflammation [118].

These works suggested that *G. lucidum* involved intrinsic immunological mechanism in the anti-tumor effect. Further studies confirmed that endogenous immunological mechanism play an important role in the anti-tumor effect of *G. lucidum* by serologic pharmacological method. After treatment with water extract from *G. lucidum* or *G. lucidum* polysaccharides by injection or oral infusion, mice serum was collected and tested. The results demonstrated that *in vitro G. lucidum* extract-treated serum inhibited proliferation of S-180 tumor cells and induced their apoptosis. Meanwhile, TNF-α and IFN-γ level of *G. lucidum* extract-treated serum significantly increased [86,92]. Similarly, *G. lucidum* polysaccharides B-treated serum had the same effects in HL-60 cells [91,96,103]. Addition of *G. lucidum* polysaccharides to the culture of macrophage and T lymphocyte promoted TNF-α and IFN-γ production and their mRNA expression in a dose-dependent manner [86,92]. Mononuclear cell conditioned media with *G. lucidum* polysaccharides (PSG-MNC-CM) containing TNF-α and IFN-γ suppressed the proliferation clonogenicity of both the HL-60 and the U937 leukemic cell lines, induced their apoptosis and triggered their differentiation. But either *G. lucidum* polysaccharides alone or normal mononuclear cell-conditioned media without *G. lucidum* polysaccharides (MNC-CM) had no such effects even at higher dose of 400 mg/ml [102]. Antibody neutralization studies confirmed that adding antibody of anti-TNF-α or anti-IFN-γ in the *G. lucidum*-treated serum markedly antagonize tumor-inhibiting effect of the *G. lucidum*-treated serum [119]. These findings revealed that *G. lucidum* inhibited tumor cells proliferation and induced their apoptosis via enhancing endogenous immune system function such as secreting anti-tumor cytokines TNF-α and IFN-γ.

G. lucidum polysaccharides synergize cytokines to induce immunological effector cells. Some reports demonstrated that *G. lucidum* polysaccharides could enhance the cytotoxicity of CTL and natural killer (NK) cells in mice and lymphokine-activated killer (LAK) cells derive from human cord blood [120–121]. Cytokine-induced killer (CIK) cells have been shown to generate effector cells with higher proliferative capacity, increased cytotoxicity and fewer side effects than LAK cells [122]. By synergizing cytokines, *G. lucidum* polysaccharides (400 mg/l or 100 mg/l) could decrease the amount of cytokine in lymphokine-activated killer (LAK) cells and CIK cells culture, but had no significant effect on the proliferation, cytotoxicity or phenotype of LAK cells and CIK cells induced by cytokines in higher doses alone. The activity of *G. lucidum* polysaccharides was relevant to enhancing IL-2, TNF production, protein and mRNA expression of granuzyme B and perforin in CIK cells, and mostly be blocked by anti-CR3 (complement receptor type 3), which suggested that the effect of *G. lucidum* polysaccharides on CIK cells was

possibly mediated primarily through complement receptor type 3 (CR3) [123–124].

Several other authors also reported on enhancement of cytokine production, both from macrophages and T lymphocytes, especially tumor necrosis factor-α (TNF-α) and interferon-γ (IFN-γ).

Both *G. lucidum* and *G. lucidum* polysaccharides promoted proliferation of lymphocytes induced by concanavalin A or lipopolysaccharide, and potentiated interlukins (IL-1, IL-2, IL-3 and IL-6), tumor necrosis factor α (TNF-α) and interferon γ (IFN-γ) production and their mRNA expression in T lymphocyte and macrophage [86,92,102,115,125].

Chen *et al.* [126] studied the effect of *G. lucidum* polysaccharides on cytokine expression in mouse splenocytes. One of the fractions (F3) has activated the expression of IL-1, IL-6, IL-12, IFN-γ, TNF-α, GM-CSF, G-CSF and M-CSF. Together with previous studies, the mode of action on macrophages was proposed, where F3 binds to TLR4 receptor and activates extracellular signal-regulated kinase, c-Jun N-terminal kinase and p38 to induce IL-1 expression.

Boh *et al.* [104] reported on *in vitro* testing of *G. lucidum* extracellular and intercellular polysaccharide fractions on the induction of cytokine synthesis in primary cultures of human mononuclear cells from a buffy coat of healthy donors. Water-soluble polysaccharides, isolated from the mycelia produced by submerged cultivation in a stirred tank bioreactor, induced 3.0–630 pg/ml of TNF-α and 1.23–2.18 pg/ml of IFN-γ.

Zhu and Lin [124] studied the interaction between *G. lucidum* polysaccharides (*Gl*-PS) and cytokines, and explored mechanisms of *Gl*-PS acting on proliferation and anti-tumor activity of cytokine-induced killer (CIK) cells. The results suggested that 400 μg/ml or 100 μg/ml *of Gl*-PS promoted CIK cells proliferation and cytotoxicity were relevant to enhancing IL-2, TNF production, protein and mRNA expression of granzyme B and perforin in CIK cells through synergizing cytokines in decreasing doses of IL-2 and anti-CD3 by 75% and 50%.

Wang *et al.* [128] screened various *G. lucidum* strains and studied their anti-tumor and immunostimulation properties. The research on anti-proliferation of tumor cells of ethano1 extract and immunostimulation of water extract of *G. lucidum* was carrried out using K562 cells and macrophages. The results showed that tumor cells were inhibited by the ethanol extract, and the macrophages were activated by the water extract of *G. lucidum.*

In the work of Zhang *et al.* [129] polysaccharide fractions from *G. lucidum* strains significantly inhibited the proliferation of leukemia cells. Effects of fractions of fruit body on stimulating the proliferation of spleen lymphocyte and T and B cells and activating the NK activity was stronger than that of fractions of mycelia. Fractions of fruit body and mycelia had similar effect on stimulating the proliferation of T and B cells in peripheral blood mononuclear cells. Fractions of fruit body and mycelia induced peripheral blood

mononuclear cells to release TNF-α in dose-dependent manner. At low concentration fractions of mycelia had similar capacity to induce the production of TNF-α with fractions of fruit body; however, at high concentration, fractions of mycelia were better than that of fruit body.

The above studies confirmed that water extracts of *G. lucidum* and *G. lucidum* polysaccharides possess anti-tumor activity *in vivo*, but have no direct cytotoxic effect on tumor cells, which indicates that they exert anti-tumor activity mediated by intrinsic immunological mechanism involved in their immuno-modulatory action such as enhancing anti-tumor function of DC, CTL, promoting anti-tumor cytokines production and potentiating cytokines activity.

Anti-tumor effects through immonomodulation and anti-angiogenesis

Some polysaccharides and peptides of *G. lucidum* have shown anti-tumor effects by inhibiting angiogenesis. Investigations revealed that administrating intragastrically *G. lucidum* polysaccharides 50, 100, 200 mg/kg markedly inhibited xenograft (human lung carcinoma cell PG) in BALB/c immune-default nude mice *in vivo*, and the *G. lucidum* polysaccharides-treated serum potently inhibited PG cell proliferation but not *G. lucidum* polysaccharides 0.1–100 mg/l alone *in vitro*. Because BALB/c nude mice have congenital T lymphocyte deficiency, and above *G. lucidum* polysaccharides could not affect macrophages function, there may be other anti-tumor mechanisms of *G. lucidum* polysaccharides except immunological mechanisms. A study by Cao and Lin [130,131] found out that *G. lucidum* polysaccharides and *G. lucidum* polysaccharides-treated serum have anti-angiogenic effect on chick chorioallantoic membrane. Further research confirmed that *G. lucidum* polysaccharides inhibited proliferation of human umbilical cord vascular endothelial (HUVEC) in a dose-dependent fashion, and decreased vascular endothelial growth factor (VEGF) secretion in human lung cancer cells in hypoxia, so anti-angiogenic effect may be the new anti-tumor mechanism of *G. lucidum* polysaccharides. Kimura *et al.* [76] found that the triterpenoid fraction of the fruit bodies of *G. lucidum* at the concentration of 800 mg/l inhibited angiogenesis induced by Matrigel – a soluble basement membrane extract of Engelbreth–Holm–Swam(EHS) tumor – supplemented with vascular endothelial growth factor (VEGF) and heparin in an *in vivo* model. Song *et al.* [132] studied anti-angiogenic and inhibitory activity on inducible nitric oxide production of the 70% ethanol extract of *G. lucidum* fresh fruiting bodies. The extract showed significant anti-angiogenic activity, which was detected using a chick embryo chorioallantoic membrane assay. Stanley *et al.* [133] examined the effect of *G. lucidum* on angiogenesis related to prostate cancer and found that *G. lucidum* inhibited the early event in angiogenesis, capillary morphogenesis of the human aortic endothelial cells. Cao and Lin [131] reported that *G. lucidum* polysaccharides peptide inhibited the growth

of vascular endothelial cell and the induction of vascular endothelial growth factor (VEGF) in human lung cancer cells. The proliferation of human umbilical cord vascular endothelial cell (HUVEC) culture was inhibited by *Gl*-PP in a dose-dependent fashion, but not because of cytotoxicity. Flow cytometric studies revealed that *Gl*-PP treatment of HUVECs could induce cell apoptosis directly. Therefore, inducing cell apoptosis by *Gl*-PP might be the mechanism of inhibiting HUVEC proliferation. Human lung carcinoma cells PG when exposed to high dose of *Gl*-PP in hypoxia for 18 h resulted in a decrease in the secreted VEGF. These findings supported the hypothesis that the key attribute of the anti-angiogenic potential of *Gl*-PP is that it may directly inhibit vascular endothelial cell proliferation or indirectly decrease growth factor expression of tumor cells.

Regulation and protection of cells

Several works suggested that *G. lucidum* also positively affects and protects living cells. Cao and Lin [106,109] found out that *G. lucidum* polysaccharides had an effect on regulation of maturation and function of dendritic cells. You and Lin *et al.* [134] reported on protective effects of *G. lucidum* glycopeptides on injury of macrophages induced by reactive oxygen species. Shi *et al.* [135] studied aqueous extracts of eight mushroom species and found out that *G. lucidum* had a potential for protecting cellular DNA from oxidative damage. Zhang *et al.* [136] reported on *in vitro* and *in vivo* protective effect of *G. lucidum* polysaccharides on pancreatic islets damage induced by alloxan. The effect was dose-dependent and visible as increased serum insulin and reduced serum glucose levels when pretreated intragastrically for 10 days in alloxan-induced diabetic mice. The mechanism was based on polysaccharide scavenging ability to protect the pancreatic islets from free radicals. It was found that the pancreas homogenates had higher lipid peroxidation products in alloxan-treated mice than in animals treated with G. lucidum polysaccharides. In a study by Lakshmi *et al.* [137] methanolic extract of *G. lucidum* had a dose-dependent protective effect on hepatic cells damage, caused by benzo[*a*]pyrene, and a significant anti-mutagenic activity *in vivo* in rats by restoring anti-oxidant defense.

Peptidoglycanes and proteins

One of the oldest known proteins isolated from *G. lucidum* is LZ-8, for which immunomodulatory and immunosuppressive activities were reported [138]. From *G. lucidum* mycelia produced by submerged fermentation, Tian and Zhang [139] purified and characterized a proteinase A inhibitor with a molecular mass of 38 kDa. The purification was carried out by ethanol precipitation (50–80%), ACA44 gel filtration and Source 30Q anion

exchange. Its carbohydrate content was about 70%. The linkage between the glycan and the core protein backbone might be O-linkage. By investigating the interaction between the inhibitor and a variety of proteinases, it was indicated that the inhibitor was more specific against yeast proteinase A than other proteinases. The inhibitor showed a remarkable heat stability.

A bioactive fraction (GLPG) was extracted and purified from the mycelia of *G. lucidum* by EtOH precipitation and DEAE-cellulose column chromatography by Liu *et al.* [140]. GLPG was a proteoglycan with a carbohydrate:protein ratio of 10.4:1. The product had anti-virus effects. Possible mode of action of antitherpetic activities of a proteoglycan isolated from the mycelia of *G. lucidum* was studied *in vitro*. Its antiviral activities against herpes simplex virus type 1 and type 2 were investigated by the cytopathic effect inhibition assay in cell culture. Although the precise mechanism was not defined yet, the work suggested that GLPG inhibited viral replication by interfering with the early events of viral adsorption and entry into target cells. Thus, the GLPG proteoglycan is a potential candidate for anti-herpes simplex virus agents.

A ribonuclease with a molecular mass of 42 kDa and with an N-terminal sequence distinct from other mushroom ribonucleases was isolated from fresh fruiting bodies of *G. lucidum* by Wang *et al.* [141]. In the purification process, the ribonuclease was adsorbed on DEAE-cellulose and Q-Sepharose, and unadsorbed on CM-Sepharose. The optimum pH of 4.0 was low compared with those reported for other mushroom ribonucleases. A temperature of 60°C was required for optimal enzyme activity. The ribonuclease was unique among mushroom ribonucleases in that it exhibited the highest potency toward poly(U), followed by poly(A). Its activity toward poly(G) and poly(C) was about one-half of that toward poly(A) and one-quarter of that toward poly(U).

Wang *et al.* [142] isolated a 15-kDa protein from fruiting bodies of *G. lucidum*, named ganodermin. The isolation procedure utilized chromatography on DEAE-cellulose, Affi-gel blue gel, CM-Sepharose and Superdex 75. Ganodermin expressed an anti-fungal activity by inhibiting the mycelial growth of *Botrytis cinerea*, *Fusarium oxysporum* and *Physalospora piricola*.

Other compounds

Polysaccharides and triterpenes have been most thoroughly investigated from *G. lucidum* and related species. However, other active compounds have also been described, such as adenosine with anti-platelet aggregation effect, lectins with mitogenic effect, alkaloids, fatty acids, vitamins and essential minerals. A structured list of chemical constituents of *G. lucidum* was given in [87].

Formulations, market products and clinical trials

Several formulations have been developed, patented and used as nutraceuticals, nutriceuticals and pharmaceuticals [143], mainly with *Ganoderma* fruiting bodies, spores and their water or ethanol extracts, rarely with purified active compounds.

Several products have undergone clinical trials and became available commercially as a syrup, injection, tablet, tincture, or bolus of powdered medicine and additives [144]. For example, Zhang and Li [145] reported on a clinical investigation of Green Valley Lingzhi capsule in treatment of 130 patients suffering from type 2 diabetes mellitus. After 2 months of a treatment, the formulation showed a synergistic effect in hypoglycemic action combined with regular hypoglycemic drugs, and significantly decreased the clinical symptoms, compared to a control group treated only with regular hypoglycemic drugs.

Shi and Qing [146] published the results of a clinical observation assessment of 547 medium and late phases cancer patients treated by a formulation Chinese *G. lucidum* Essence. The study showed that the death rate of the patients in the long-term treatment was significantly lower. A continuous 2–3-month active treatment with a daily dosage of 4–6 g of *G. lucidum* essence was proposed, with further dosage of 2 g/day continuously after the third month of therapy. Short-term treatments were less successful.

Sliva *et al.* [147] reported that *G. lucidum* spores and unpurified fruiting body inhibited invasion of breast and prostate cancer cells by a common mechanism and could have potential therapeutic use for cancer treatment. In their study, they investigated the effect of *G. lucidum* on highly invasive breast and prostate cancer cells. Spores or dried fruiting body inhibited constitutively active transcription factors AP-1 and NF-jB in breast MDA-MB-231 and prostate PC-3 cancer cells. Furthermore, *Ganoderma* inhibition of expression of uPA and uPA receptor (uPAR), as well as secretion of uPA, resulted in the suppression of the migration of MDA-MB-231 and PC-3 cells.

Noguchi *et al.* [148] performed a phase I study of a methanol extract of *G. lucidum* in men with mild symptoms of bladder outlet obstruction. Male volunteers of age >50 were enrolled. The overall administration was well tolerated with no adverse effects. Statistically significant reductions in International Prostate Symptoms Score (I-PSS) versus placebo were observed at the 6 mg and 60 mg dose. The study concluded that the extract of *G. lucidum* was well tolerated, a significant improvement in I-PSS was observed and a 6 mg dose of the extract was recommended for the phase II trial.

A study by Chen *et al.* [149] evaluated the effects of *G. lucidum* polysaccharides on patients with advanced colorectal cancer. Forty-seven patients were enrolled and treated with oral *G. lucidum* at 5.4 g/day for 12 weeks. Selected immune parameters were monitored using various immunological methods throughout the study. In 41 assessable cancer patients, treatment

with *G. lucidum* tended to increase mitogenic reactivity to phytohemagglutinin, counts of CD3, CD4, CD8 and CD56 lymphocytes, plasma concentrations of interleukin (IL)-2, IL-6 and interferon (IFN)-γ, and NK activity, whereas plasma concentrations of IL-1 and tumor necrosis factor (TNF)-α were decreased. For all of these parameters, no statistical significance was observed when a comparison was conducted between baseline and those values after a 12-week treatment with *G. lucidum*. The changes of IL-1 were correlated with those for IL-6, IFN-γ, CD3, CD4, CD8 and NK activity ($p < 0.05$) and IL-2 changes were correlated with those for IL-6, CD8 and NK activity. The results indicated that *G. lucidum* may have potential immunomodulating effect in patients with advanced colorectal cancer; however, further studies are needed to explore the benefits and safety of *G. lucidum* in cancer patients.

Clinical studies also showed that G. *lucidum* preparations exerted synergistic therapeutic effect when used in conjunction with radiation and chemotherapy, reduced the following side effects: leukopenia, thrombocytopenia, anemia, nausea, vomiting, appetite loss, anti-infection deficiency and immunosuppression, enhanced tolerance for radiation and chemotherapy, constitution and immunity in cancer patients to potentiate the therapeutic efficacy and ameliorate adverse toxicity of radiation and chemotherapy [149–153]. The therapeutic efficacy of *G. lucidum* on cancer patients is mediated not only by immuno-modulation, such as enhancing function of DC, CTL and other immunological effector cells to kill tumor cells, and promoting anti-tumor cytokines production and activity, but also by anti-angiogenesis. *G. lucidum* abates or avoids the toxicity caused by other therapies via stimulating hematopoiesis and antagonizing damage resulted from radiation and chemotherapy. It is obvious that the effect of *G. lucidum* on radiation and chemotherapy exactly compensates for the deficit of the two therapies by strengthening internal immune function to resist external malign factors caused by tumor or toxicity of radiation and chemotherapy.

Although preparations of *G. lucidum* have been commonly used in China as self-medication for the prevention and treatment of various medical diseases including cancer, liver diseases, hypertension, hyperlipidemia and coronary disease, there were, until recently, no reports on their side effects. However, in 2004, Yuen *et al.* [127] reported on a case of hepatotoxicity, probably due to a formulation of *G. lucidum*. A patient was a 78-year-old Chinese lady in whom an intake of a formulation of *G. lucidum* powder caused a significant hepatotoxicity in which the liver biochemistry mimiced that of acute cholangitis. The patient took regular self-medications including oral calcium and multivitamin tablets daily. She also had regular intake of *G. lucidum* as a health supplement for at least 1 year, but she started to take a new powder formulation 4 weeks before the onset of symptoms. The liver toxicity was therefore most likely due to the ingredients of the powder *G. lucidum* formulation.

Conclusion

Reports on isolated compounds from *G. lucidum* are very convincing; there is abundant evidence that triterpenoids, polysaccharides and proteoglycans are effective. In most cases, extracts of partly-purified preparations have been used for *in vitro* or *in vivo* testing. Synergistic effects of mixtures of active components have been known; however, their biological activities need further assessment before they can be accepted not only by the traditional Asian medicine, but also by the Western science and medicine. Modern biotechnological cultivation methods in bioreactors enable fast, efficient and economical production of *G. lucidum* biomass in sufficient quantities for potential future pharmaceutical industrial production.

References

1. Leung SWS. Lingzhi (*Ganoderma*) research – the past, present and future perspectives. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on Ganoderma Research, Shanghai, October 21–23, Beijing, Medical University Press, 2002, pp. 1–9.
2. Kim HW and Kim BK. Recent advances on the biologically active triterpenoids of *Ganoderma lucidum*. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on Ganoderma Research, Shanghai, October 21–23, Beijing, Medical University Press, 2002, pp. 10–19.
3. Jong SC and Birmingham JM. Medicinal benefits of the mushrooms *Ganoderma*. Adv Appl Microbiol 1992;37:101–134.
4. Tasaka K, Akagi M, Miyoshi K, Mio M and Makino T. Anti-allergic constituents in the culture medium of *Ganoderma lucidum*. 1. Inhibitory effect of oleic acid on histamine release. Agents Action 1988;23:153–156.
5. Kino K, Sone T, Watanabe J, Yamashita A, Tsuboi H, Miyajima H and Tsunoo H. Immunomodulator, LZ-8, prevents antibody-production in mice. Int J Immunopharmacol 1991;13:1109–1115.
6. van Der Hem LG, Van Der Vliet J A, Bocken C F M, Kino K, Hoitsma A J and Tax WJM. Ling zhi-8 – studies of a new immunomodulating agent. Transplantation (Baltimore) 1995;60:438–443.
7. Maruyama H, Yamazaki K, Murofushi S, Konda C and Ikekawa T. Antitumor activity of *Sarcodon aspratus* (Berk.) S. Ito and *Ganoderma lucidum* (Fr.) Karst. J Pharmacobio-Dyn 1989;12:118–123.
8. Lee SY and Rhee H M. Cardiovascular effects of mycelium extract of *Ganoderma lucidum* – inhibition of sympathetic outflow as a mechanism of its hypotensive action. Chem Pharm Bull 1990;38:1359–1364.
9. Liu GT, Bao X, Niu S, Li Z and Sung Z. Some pharmacological actions of the spores of *Ganoderma lucidum* and the mycelium of *Ganoderma capense* (Lloyd) Teng cultivated by submerged fermentation. Chin Med J 1979;92:469–500.
10. Hirose K, Muto S, Niimura K, Ohara M, Oguchi Y, Matsunaga K, Kadochi J, Sugita N, Furushu T, Yoshikumi C and Takahashi M. Antiviral agent. Japanese Patent JP 63316734, 1988.

11. Mizumoto K, Yamashita A, Kii M and Sumio H. Antiretrovirus agent. Japanese Patent JP 2032026, 1990.
12. Kim HW, Shim MJ, Choi EC and Kim BK. Inhibition of cytopathic effect of human immunodeficiency virus-1 by water-soluble extract of *Ganoderma lucidum.* Arch Pharm Res 1997;20:425–431.
13. Hattori M, El-Mekkawy S and Meselhy R. Inhibitory effects of components from *Ganoderma lucidum* on the growth of human immunodeficiency virus (HIV) and the protease activity. In: Proceedings of the 1st International Symposium on *Ganoderma Lucidum* in Japan, Mizuno T, Ide N and Hasegawa Y (eds), November 17–18, 1997, pp. 128–135.
14. El-Mekkawy S, Meselhy M R, Nakamura N, Tezuka Y, Hattori M, Kakiuchi N, Shimotohno K, Kawahata T and Otake T. Anti-HIV-1 and anti-HIV-1 protease substances from *Ganoderma lucidum*. Phytochemistry 1998;49:1651–1657.
15. Min BS, Nakamura N, Miyashiro H, Bae K W and Hattori M. Triterpenes from the spores of *Ganoderma lucidum* and their inhibitory activity against HIV-1 protease. Chem Pharm Bull 1998;46:1607–1612.
16. Triratana S, Thaithatgoon S and Gawgla M. Cultivation of *Ganoderma lucidum* in sawdust bags. In: Science and Cultivation of Edible Fungi, Maher MJ (ed), Proceedings of the 13th International Congress on the Science Cultivation of Edible Fungi, Dublin, September 1–6, 1991, Rotterdam, A. A. Balkema, pp. 567–572.
17. Kohlmünzer S, Wegiel J and Sitarz J. Polysaccharides in mycelial culture of *Ganoderma applanatum* (Pers.) Pat. Herba Hung 1989;28:87–94.
18. Lin SQ, Wang SZ, Lin SG and Lin ZB. Studies on *Ganoderma* submerged fermentation and its product extraction technique. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 93–97.
19. Habijanic J and Berovic M. The relevance of solid-state substrate moisturing on *Ganoderma lucidum* biomass cultivation. Food Technol Biotechnol 2000;38:225–228.
20. Berovic M, Habijanic J, Zore I, Wraber B, Hodzar D, Boh B and Pohleven F. Submerged cultivation of *Ganoderma lucidum* biomass and immunostimulatory effects of fungal polysaccharides. J Biotechnol 2003;103:77–86.
21. Mizuno T, Wang G, Zhang J, Kawagishi H, Nishitoba T and Li J. Reishi, *Ganoderma lucidum* and *Ganoderma tsugae*: bioactive substance and medicinal effects. Food Rev Int 1995;11:151–166.
22. Miyahara R, Yoshimoto T and Asawa K. Chemical structures and changes of extracts during growth of reishi (*Ganoderma lucidum*). Mokuzai Gakkaishi 1987;33:416–422.
23. Chan K. Linking chemical and biological characteristics in assuring the quality of Chinese medicinal materials and OTC products. In: L *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 24–34.
24. Lee JH. Cultivation of reishi (*Ganoerma lucidum*) [online]. MushWorld – Cultivation, 2000-12-24, available at http://www.mushworld.com/sub_en.html
25. Chen AW. A fresh look at an ancient mushroom *Ganoderma lucidum* (Reishi) [online]. MushWorld – Cultivation, 2003-03-18, available at http://www.mushworld.com/sub_en.html
26. Chen AW. Natural-log cultivation of the medicinal mushroom *Ganoderma lucidum* (Reishi) [online]. MushWorld – Cultivation, 2004-01-09, available at http://www.mushworld.com/sub_en.html

27. Sukarno N. Development of *Ganoderma lucidum* on soft and hard wood logs and determination of organic germanium and ganoderic acid content of the fruiting body produced [online]. MushWorld – Cultivation, 2004-10-06, available at http://www.mushworld.com/sub_en.html
28. Chen AW. Cultivation of the medicinal mushroom *Ganoderma lucidum* (Curt.: Fr.) P. Karst (Reishi) in North America (3) [online]. MushWorld – Cultivation, 2002-02-01, available at http://www.mushworld.com/sub_en.html
29. Royse DJ. Specialty mushrooms. In: Progress in New Crops, Janick J (ed), Arlington, VA, ASHS Press, 1996, pp. 464–475.
30. Chen AW. Growing *Ganoderma* mushrooms. In: Mushroom Grower's Handbook I, Part III: Mushrooms Worldwide, Chapter 11: Mushrooms for the tropics. ISSN 1739-1377, [online]. MushWorld, 2004, pp. 224–234, available at http://www.mushworld.com/service/handbook/english/eng-book1/chapter11-01_p.224.pdf
31. Yukinori S, Tetsuji S and Yasushi S. Cultivation of *Ganoderma lucidum* Karst. Patent JP11146728, JMC KK, 1999-06-02.
32. Furci George-Nascimento GM. *Ganoderma lucidum* (Curt.:Fr.) P. Karst grown indoors on native Chilean hardwoods [online]. MushWorld – Cultivation, 2005-07-30, available at http://www.mushworld.com/sub_en.html
33. Gonzalez-Matute R, Figlas D, Devalis R, Delmastro S and Curvetto N. Sunflower seed hulls as a main nutrient source for cultivating *Ganoderma lucidum*. Micologia Aplicada Int. 2002;14:19–24.
34. Yang FC, Hsieh C and Chen HM. Use of stillage grain from a rice-spirit distillery in the solid state fermentation of *Ganoderma lucidum*. Process Biochem 2003;39:21–26.
35. Hsieh C and Yang FC. Reusing soy residue for the solid-state fermentation of *Ganoderma lucidum*. Bioresour Technol 2004;91:105–109.
36. Kim HK. Comparison of characteristics of *Ganoderma lucidum* according to geographical origins: consideration of growth characteristics (1) [online]. MushWorld – World Mushroom, 2001-09-01, available at http://www.mushworld.com/sub_en.html
37. Takashi M. Apparatus for cultivating *Ganoderma lucidum* Karst and other mushrooms and their cultivation. Patent JP11155366, 1999.
38. Kiyoshi K, Hamajirou S and Yoshiaki A. Cultivation method for *Ganoderma lucidum* (fr.) Karst. Patent JP10084772, 1998.
39. Shigeru Y. Cultivation of Reishi. Patent JP3083521, 1991.
40. Habjanic J and Berovic M. Process of cultivation of fungus Ganoderma lucidum on a solid cultivation substrate. Patent SI 20923, 2002.
41. Li PJ and Shen CG. Method for propagating fungi using solid state fermentation. Patent WO0220727, 2002.
42. Chang MY, Tsai GJ and Houng JY. Optimization of the medium composition for the submerged culture of *Ganoderma lucidum* by Taguchi array design and steepest ascent method. Enzyme Microb Technol 2006;38:407–414.
43. Hsieh C, Tseng MH and Liu CJ. Production of polysaccharides from *Ganoderma lucidum* (CCRC 36041) under limitations of nutrients. Enzyme Microb Technol 2006;38:109–117.
44. Zore I, Berovic M, Boh B, Hodzar D and Pohleven F. Procedure for preparation of inoculum for growing of fungus *Ganoderma lucidum* by submersion fermentation. Patent SI 9700014, 1998.
45. Yang FC and Liau CB. The influence of environmental conditions on polysaccharide formation by *Ganoderma lucidum* in submerged cultures. Process Biochem 1998;33:547–553.

46. Lee KM, Lee SY and Lee HY. Bistage control of pH for improving exopolysaccharide production from mycelia of *Ganoderma lucidum* in an air-lift fermentor. J Biosci Bioeng 1999;88:646–650.
47. Fang QH and Zhong JJ. Effect of initial pH on production of ganoderic acid and polysaccharide by submerged fermentation of *Ganoderma lucidum*. Process Biochem 2002;37:769–774.
48. Fang QH and Zhong JJ. Submerged fermentation of higher fungus *Ganoderma lucidum* for production of valuable bioactive metabolites – ganoderic acid and polysaccharide. Biochem Eng J 2002;10:61–65.
49. Fang QH, Tang Yj and Zhong JJ. Significance of inoculation density control in production of polysaccharide and ganoderic acid by submerged culture of *Ganoderma lucidum*. Process Biochem 2002;37:1375–1379.
50. Tang YJ and Zhong JJ. Fed-batch fermentation of *Ganoderma lucidum* for hyperproduction of polysaccharide and ganoderic acid. Enzyme Microb Tech 2002;31:20–28.
51. Tang YJ and Zhong JJ. Role of oxygen supply in submerged fermentation of *Ganoderma lucidum* for production of *Ganoderma* polysaccharide and ganoderic acid. Enzyme Microb Technol 2003;32:478–484.
52. Hsieh C, Hsu TH and Yang FC. Production of polysaccharides of *Ganoderma lucidum* (CCRC36021) by reusing thin stillage. Process Biochem 2005;40:909–916.
53. Lee H, Song M, Yu Y and Hwang S. Production of *Ganoderma lucidum* mycelium using cheese whey as an alternative substrate: response surface analysis and biokinetics. Biochem Eng J 2003;15:93–99.
54. Lee H, Song M, Yu Y and Hwang S. Optimizing bioconversion of deproteinated cheese whey to mycelia of *Ganoderma lucidum*. Process Biochem 2003;38:1685–1693.
55. Hirotani M, Ino C, Furuya T and Shiro M. Ganoderic acids T, S, and R, new triterpenoids from the cultured media of *Ganoderma lucidum*. Chem Pharm Bull 1986;34:2282–2285.
56. Kikuchi T, Kanomi S, Murai Y, Kadota S, Tsubono K and Ogita Z. Constituents of the fungus *Ganoderma lucidum* (Fr.) Karst. I. Structures of ganoderic acids C2, E, I, and K, lucidenic acid F and related compounds. Chem Pharm Bull 1986;34: 3695–3712.
57. Morigiwa A, Kitabatake K, Fujimoto Y and Ikekawa N. Angiotensin converting enzyme inhibitory triterpenes from *Ganoderma lucidum*. Chem Pharm Bull 1986;34:3025–3028.
58. Nishitoba T, Sato H and Sakamura S. Triterpenoids from the fungus *Ganoderma lucidum*. Phytochemistry 1987;26:1777–1784.
59. Nishitoba T, Goto S, Sato H and Sakamura S. Bitter triterpenoids from the fungus *Ganoderma applanatum*. Phytochemistry 1989;28:193–197.
60. Lin LJ and Shiao MS. Seven new triterpenes from *Ganoderma lucidum*. J Nat Prod 1988;51:918–924.
61. Lin LJ, Shiao MS and Yeh SF. Triterpenes from *Ganoderma lucidum*. Phytochem 1988;27:2269–2271.
62. Shiao MS, Lin LJ, Yeh SF and Chou CS. Two new triterpenes of the fungus *Ganoderma lucidum*. J Nat Prod 1987;50:886–890.
63. Chairul, Tokuyama T, Hayashi Y, Nishizawa M, Tokuda H, Chairul SM and Hayashi Y. Applanoxidic acids A, B, C and D, biologically active tetracyclic triterpenes from *Ganoderma applanatum*. Phytochem 1991;30:4105–4109.
64. Chairul, Hayashi Y and Chairul SM. Lanostanoid triterpenes from *Ganoderma applanatum*. Phytochem 1994;35:1305–1308.

65. Boh B, Hodzar D, Dolnicar D, Berovic M and Pohleven F. Isolation and quantification of triterpenoid acids from *Ganoderma applanatum* of Istrian origin. Food Technol Biotechnol 2000;1:11–18.
66. Kim DH, Shim SB, Kim NJ and Jang IS. Beta-glucuronidase-inhibitory and hepatoprotective effect of *Ganoderma lucidum*. Biol Pharm Bull 1999;22:162–164.
67. Toth JO, Luu B and Ourisson G. Les acides ganoderiques T a Z: triterpenes cytotoxiques de *Ganoderma lucidum* (Polyporaceae). Tetrahedron Lett 1983;24: 1081–1084.
68. Lin SB, Li CH, Chen YR, Kan LS and Lee SS. Triterpene extract from *Ganoderma lucidum* inhibits growth of hepatoma Huh7 cells: involvement of oxidative stress induction. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 176–182.
69. Gao JJ, Min BS, Ahn EM, Nakamura N, Lee HK and Hattori M. New triterpene aldehydes, lucialdehydes A-C, from *Ganoderma lucidum* and their cytotoxicity against murine and human tumor cells. Chem Pharm Bull (Tokyo) 2002;50:837–840.
70. Min BS, Gao JJ, Nakamura N and Hattori M. Triterpenes from the spores of *Ganoderma lucidum* and their cytotoxicity against meth-A and LLC tumor cells. Chem Pharm Bull (Tokyo) 2000;48:1026–1033.
71. Lin SB, Li CH, Lee SS and Kan LS. Triterpene-enriched extracts from *Ganoderma lucidum* inhibit growth of hepatoma cells via suppressing protein kinase C, activating mitogen-activated protein kinases and G2-phase cell cycle arrest. Life Sci 2003;72:2381–2390.
72. Zhu HS, Yang XL, Wang LB, Zhao DX and Chen L. Effects of extracts from sporoderm-broken spores of *Ganoderma lucidum* on HeLa cells. Cell Biol Toxicol 2000;16:201–206.
73. Yu SP, Zhang JS, Tang QJ, Shi XM, Liu YF, Yan Y and Pan YL. Correlation between intracellular triterpenes from mycelia of *Ganoderma lucidum* in different growth stages and inhibition effect on tumor cells. Mycosystema 2004;23:548–554.
74. Liu J, Shimizu K, Konishi F, Noda K, Kumamoto S, Kurashiki K and Kondo R. Antiandrogenic activities of the triterpenoids fraction of *Ganoderma lucidum*. Food Chem 2007;100:1691–1696.
75. Mueller CI, Kumagai T, Kelly JO, Seeram NP, Heber D and Koeffler HP. *Ganoderma lucidum* causes apoptosis in leukemia, lymphoma and multiple myeloma cells. Leukemia Res 2006;30:841–848.
76. Kimura Y, Taniguchi M and Baba K. Antitumor and antimetastatic effects on liver of triterpenoid fractions of *Ganoderma lucidum*: mechanism of action and isolation of an active substance. Anticancer Res 2002;22(6A):3309–3318.
77. Kabir Y, Kimura S and Tamura T. Dietary effect of *Ganoderma lucidum* mushroom on blood pressure and lipid levels in spontaneously hipertensive rats. J Nat Sci Vitaminol 1988;34:433–438.
78. Kohda H, Tokumoto W, Sakamoto K, Fujii M, Hirai Y, Yamasaki K, Komoda Y, Nakamura H, Ishihara S and Uchida M. The biologically active constituents of *Ganoderma lucidum* (Fr.) Karst histamine release-inhibitory triterpenes. Chem Pharm Bull 1985;33:1367–1374.
79. Wang CN, Chen JC, Shiao MS and Wang CT. The aggregation of human platelet induced by ganodermic acid S. Biochim Biophys Acta 1989;986:151–160.
80. Wang CN, Chen JC, Shiao MS and Wang CT. The inhibition of human platelet function by ganodermic acids. Biochem J 1991;277:189–197.

81. Su CY, Shiao MS and Wang CT. Predominant inhibition of ganodermic acid S on the thromboxane A2 -dependent pathway in human platelets response to collagen. Biochim Biophys Acta 1999;1437:223–234.
82. Su C, Shiao M and Wang C. Potentiation of ganoderemic acid S on prostaglandin E(1)-induced cyclic AMP elevation in human platelets. Thromb Res 2000;99:135–145.
83. Min BS, Gao JJ, Hattori M, Lee HK and Kim YH. Anticomplement activity of triterpenoids from the spores of *Ganoderma lucidum*. Planta Med 2001;67:811–814.
84. Fang JN, Bao XF and Yuen WH. Studies on the polysaccharides from spores of *Ganoderma lucidum.* In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 98–103.
85. Lin SQ, Wang SZ, Lin ZB and Lin YX. Purification and identification of glycopeptides from *Ganoderma lucidum* fruit bodies cultivated with grass and wood log. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 109–114.
86. Zhang QH and Lin ZB. Effect of *Ganoderma lucidum* polysaccharides B on TNF-α and INF-γ production and their mRNA expression. J Beijing Med Univ 1999;31:179–183.
87. Russell R and Paterson M. *Ganoderma* – A therapeutic fungal biofactory. Phytochem 2006;67:1985–2001.
88. Sasaki T, Arai Y, Ikekawa T, Chihara G and Fukuoka F. Antitumor polysaccharides from some Polyporaceae, *Ganoderma applanatum* (Pers.) Pat. and *Phellinus linteus* (Berk. Et Curt) Aoshima. Chem Pharm Bull 1971;19:821–826.
89. Lee SS, Chen FD, Chang SC, Wei YH, Liu I and Chen C. In vivo antitumor effect of crude extracts from the mycelium of *Ganoderma lucidum*. Bull Chinese Oncol Soc 1984;5:22–27.
90. Hwang SF, Liu KJ, Kuan YH, Tung KS, Su CH and Tung TC. The inhibitory effect on artificial pulmonary metastasis of murine S-180 Sarcoma cells by orally administered *Ganoderma lucidum*. J Chim Oncol Soc 1989;5:10–15.
91. Furusawa E, Chou SC, Furasawa S, Hirazum A and Dang Y. Antitumor activity of *Ganoderma lucidum*, and edible mushroom, on intraperitoneally implanted Lewis lung carcinoma in synergeneic mice. Phytother Res 1992;6:300–304.
92. Zhang Q and Lin ZB. The antitumor activity of *Ganoderma lucidum* (Curt,:Fr.) P. Karst. (Lingzhi) (Aphylophoromycetidae) polysaccharides is related to tumor necrosis factor-α and interferon-γ. Int J Med Mushroom 1999;1:207–215.
93. Lu H, Kyo E, Uesaka T, Katoh O and Watanabe H. Prevention of development of *N*,*N*′-dimethylhydrazine-induced colon tumors by a water-soluble extract from cultured medium of *Ganoderma lucidum* (Rei-shi) mycelia in male ICR mice. Int J Mol Med 2002;9:113–117.
94. Liu X, Yuan JP, Chuang CK and Chen XJ. Antitumor activity of the sporoderm-broken germinating spores of *Ganoderma lucidum*. Cancer Lett 2002;182:155–161.
95. Lu H, Kyo E, Uesaka T, Katoh O and Watanabe H. A water-soluble extract from cultured medium of *Ganoderma lucidum* (Rei-shi) mycelia suppresses azoxymethane-induction of colon cancers in male F344 rats. Oncol Rep 2003;10:375–379.
96. Hu YH and Lin ZB. Polysaccharides isolated from mycelia of *Ganoderma lucidum* induced HL-60 cell apoptosis by enhancing macrophage activity. Chin Pharmacol Bull 1999;5:27–30.
97. Hu YH and Lin ZB. Effects of polysaccharides isolated from mycelia of *Ganoderma lucidum* on HL-60 cell apoptosis. Acta Pharm Sin 1999;34:268–271.

98. Zhang NQ.. Basic Theory of Traditional Chinese Medicine, 1st edn, Vol. I, Beijing, People Sanitation Press, 1990. pp. 199–205.
99. Lin ZB. Modern Research of *Ganoderma lucidum*, 2nd edn, Beijing, Beijing Medical University Press, 2001 (Chapter 1 and 8).
100. Lei LS, Li MC and Sun LS. *Ganoderma lucidum* and its components on the function of macrophages. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 20–23.
101. Shiuh S, Yau H, Chieh F, Sheng Y and Kuang Y. Antitumor effects of *Ganoderma lucidum*. J Chin Med 1995;6:1–12.
102. Wang SY, Hsu ML, Hsu HC, Tzeng CH, Lee SS, Shiao MS and Ho CK. The antitumor effect of *Ganoderma lucidum* is mediated by cytokines released from activated macrophages and T lymphocyes. Int J Cancer 1997;70:699–705.
103. Zhang QH and Lin ZB. The antitumor activity of *Ganoderma lucidum* (Curt; Fr.) P. Karst (Ling Zhi) (Aphyllophoromycetideae) polysaccharides is related to tumor necrosis factor-α and interferon-γ. Int J Med Mushroom 1999;1:207–215.
104. Boh B, Berovic M, Wraber B, Hodzar D, Habijanic J, Pohleven F and Zore I. *Ganoderma lucidum* (W.Curt.:Fr.) Lloyd and G. applanatum (Pers.) Pat. (Aphyllophoromycetideae) from Slovenian habitats: cultivation, isolation, and testing of active compounds. Int J Medic Mushroom 2004;6:15–32.
105. Lee SS, Lee PL, Chen CF, Wang SY and Chen KY. Antitumor effects of *Ganoderma lucidum*. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 46–57.
106. Cao LZ and Lin ZB. Regulation of *Ganoderma lucidum* polysaccharides on cytotoxic T lymphocytes induced by dendric cells in vitro. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 122–129.
107. Lei L, Lin Z, Chen Q, Li R and He Y. Antagonistic effect of *Ganoderma lucidum* polysaccharide on the immunosuppressive response induced by cyclosporin A, hydrocortisone and antitumor agents. Chin J Pharmacol Toxicol 1993;7:183–185.
108. Li M, Lei L and Liang D. Effect of *Ganoderma* polysaccharide on intracellular free calcium in murine peritoneal macrophages. Chin Pharm J 1999;34:805–807.
109. Cao LZ and Lin ZB. Regulation on maturation and function of dendritic cells by *Ganoderma lucidum* polysaccharides. Immunol Lett 2002;83:163–169.
110. Cao LZ and Lin ZB. Regulation effect of *Ganoderma lucidum* polysaccharides on cytotoxic T-lymphocytes induced by dendritic cells in vitro. Acta Pharmacol Sin 2003;24:321–326.
111. Shao BM, Dai H, Xu W, Lin ZB and Gao XM. Immune receptors for polysaccharides form *Ganoderma lucidum*. Biochem Bioph Res Commun 2004;323:133–141.
112. Chien CM, Cheng JL, Chang WT, Tien MH, Tsao CM, Chang YH, Chang HY, Hsieh JF, Wong CH and Chen ST. Polysaccharides of *Ganoderma lucidum* alter cell immunophenotypic expression and enhance CD56+ NK-cell cytotoxicity in cord blood. Bioorg Med Chem 2004;12:5603–5609.
113. Lei LS and Lin ZB. Effects of *Ganoderma* polysaccharides on the activity of DNA polymerase in spleen cells stimulated by alloantigents in mice in vitro. J Beijing Med Univ 1991;23:329–333.
114. Lei LS and Lin ZB. Effects of *Ganoderma* polysaccharides on the MLC reaction. Basic Med Clin 1992;12:59–60.

115. Cao LZ and Lin ZB. Comparison of the effects of polysaccharides from wood-cultured and bag-cultured *Ganoderma lucidum* on murine spleen lymphocyte proliferation in vitro. Acta Pharmacol Sin 2003;38:92–97.
116. Bao XF, Wang XS, Dong Q, Fang JN and Li XY. Structural features of immunologically active polysaccharides from *Ganoderma lucidum*. Phytochem 2002;59:175–181.
117. Zhang JS, Tang QJ, Zimmerman-Kordmann M, Reuter W and Fan H. Activation of B lymphocytes by GLIS, a bioactive proteoglycan from *Ganoderma lucidum*. Life Sci 2002;71:623–638.
118. Liu YH, Tsai CF, Kao MC, Lai YL and Tsai JJ. Effectiveness of Dp2 nasal therapy for Dp2-induced airway inflammation in mice: using oral *Ganoderma lucidum* as an immunomodulator. J Microbiol Immunol Infect 2003;36:236–242.
119. Lin ZB and Wang PY. The Pharmacological Study of *Ganoderma* Spores and Its Active Components, J Peking Univ (Health Sciences), 2006;38(5):541–547.
120. Xu X, Hou GH, Cao RH and Xie WG. Study on mechanism of *Ganodermaa lucidum* polysaccharides enhancing activity of LAK cells from human cord blood. Chinese J Cancer Biotherapy 1997;4:236.
121. Lin ZB and Zhang HN. Anti-tumor and immunoregulatory activities of *Ganoderma lucidum* and its possible mechanisms. Acta Pharmacol Sin 2004;25:1387–1391.
122. Lu PH and Negrin RS. A novel population of expended human CD3 + CD56 + cells derived from T cells with potent in vivo antitumor activity in mice with severe combined immunodeficiency. J Immunol 1994;153:1687–1696.
123. Zhu XL and Lin ZB. Effects of *Ganoderma lucidum* polysaccharides on proliferation and cytotoxicity of cytokine-induced killer cells. Acta Pharm Sin 2005;26:1130–1137.
124. Zhu XL and Lin ZB. Modulation of cytokines production, granzyme B and perforin in murine CIK cells by *Ganoderma lucidum* polysaccharides. Carbohyd Polym 2006;63:188–197.
125. Wang YY, Khoo KH, Chen ST, Lin CC, Wong CH and Lin CH. Studies on the immuno-modulating and antitumor activities of *Ganoderma lucidum* (Reishi) polysaccharides: functional and protewmic analyses of a fucose-containing glycoprotein fraction responsible for the activities. Bioorg Med Chem 2002;10:1057–1062.
126. Chen HS, Tsai YF, Lin S, Lin CC, Khoo KH, Lin CH and Wong CH. Studies on the immuno-modulating and anti-tumor activities of *Ganoderma lucidum* (Reishi) polysaccharides. Bioorg Med Chem 2004;12:5595–5601.
127. Yuen MF, Ip P, Ng WK and Lai CL. Hepatotoxicity due to a formulation of *Ganoderma lucidum* (lingzhi). Letters to the editor. J Hepatol 2004;41:685–690.
128. Wang XX, Tang QJ, Zhang JS, Yang Y, Liu YF, Jia W and Liu F. Effect of various grades of *Ganoderma lucidum* on tumor inhibition and immunostimulation. Acta Edul Fung 2005;12:48–51.
129. Zhang JS, Jia W, Xing ZT, Tang QJ, Liu YF, Yang Y, Zhou CY and Liu F. Comparison of bioactivity of fruiting body and mycelia of *Ganoderma lucidum* and their purified fractions. Mycosystema 2004;23:85–92.
130. Cao QZ and Lin ZB. Antitumor and anti-angiogenic activity of *Ganoderma lucidum* polysaccharides peptide. Acta Pharmacol Sin 2004;25:833–838.
131. Cao QZ and Lin ZB. *Ganoderma lucidum* polysaccharides peptide inhibits the growth of vascular endothelial cell and the induction of VEGF in human lung cancer cell. Life Sci 2006;78:1457–1463.
132. Song YS, Kim SH, Sa JH, Jin C, Lim CL and Park EH. Anti-angiogenic and inhibitory activity on inducible nitric oxide production of the mushroom *Ganoderma lucidum*. J Ethnopharmacol 2004;90:17–20.

133. Stanley G, Harvey K, Slivova V, Jiang J and Sliva D. *Ganoderma lucidum* suppresses angiogenesis through the inhibition of secretion of VEGF and TGF-β1 from prostate cancer cells. Biochem Biophys Res Com 2005;330:46–52.
134. You YH and Lin ZB. Protective effects of *Ganoderma lucidum* polysaccharides peptide on injury of macrophages induced by reactive oxygen species. Acta Pharmacol Sin 2002;23:878–891.
135. Shi YL, James AE, Benzie IFF and Bushwell JA. Mushroom derived preparations in the prevention of H_2O_2-induced oxidative damage to cellular DNA. Teratogen Carcin Mut 2002;22:103–111.
136. Zhang HN, He JH, Yuan L and Lin ZB. In vitro and in vivo protective effect of *Ganoderma lucidum* polysaccharides on alloxan-induced pancreatic islets damage. Life Sci 2003;73:2307–2319.
137. Lakshmi B, Ajith TA, Jose N and Janardhanan KK. Antimutagenic activity of methanolic extract of *Ganoderma lucidum* and its effect on hepatic damage caused by benzo[*a*]pyrene. J Ethnopharmacol 2006;107:297–303.
138. Van der Hem LG, van der Vliet JA, Bocken CF, Kino K, Hoitsma HJ and Tax WJ. Ling Zhi-8: studies of a new immunomodulating agent. Transplantation 1995;60:438–443.
139. Tian YP and Zhang KC. Purification and characterization of a novel proteinase A inhibitor from *Ganoderma lucidum* by submerged fermentation. Enzyme Microb Technol 2005;36:357–361.
140. Liu J, Yang F, Ye LB, Yang XJ, Timani KA, Zheng Y and Wang YH. Possible mode of action of antiherpetic activities of a proteoglycan isolated from the mycelia of *Ganoderma lucidum* in vitro. J Ethnopharmacol 2004;95:256–272.
141. Wang HX, Ng TB and Chiu SW. A distinctive ribonuclease from fresh fruiting bodies of the medicinal mushroom *Ganoderma lucidum*. Biochem Biophys Res Com 2004;314:519–522.
142. Wang HX and Ng TB. Ganodermin, an antifungal protein from fruiting bodies of the medicinal mushroom *Ganoderma lucidum*. Peptides 2006;27:27–30.
143. Chang ST and Buswell JA. Mushroom nutriceuticals. World J Microbiol Biotechnol 1996;12:473–476.
144. Wasser SP and Weiss AL. Medicinal Mushrooms – *Ganoderma lucidum*, Reishi Mushroom, Haifa, Pedeifus Publishing House, 1997.
145. Zhang CY and Li YM. Clinical investigation of Green Valley Lingzhi capsule on type 2 diabetes mellitus. In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 194–198.
146. Shi KG and Quing LH. The follow-up observation assessment of medium and late phases cancer treated by Chinese *Ganoderma lucidum* essence (CGLE). In: *Ganoderma*: Genetics, Chemistry, Pharmacology and Therapeutics, Zhi-Bin Lin (ed), Proceedings of International Symposium on *Ganoderma* Research, Shanghai, October 21–23, 2002, Beijing, Medical University Press, pp. 188–193.
147. Sliva D, Labarrere C, Slivova S, Sedlak M, Lloyd FP and Ho NW. *Ganoderma lucidum* suppresses motility of highly invasive breast and prostate cancer cells. Biochem Biophys Res Commun 2002;298:603–612.
148. Noguchi M, Kakuma T, Tomiyasu K, Konishi F, Kumamoto S, Kondo R and Matsuoka K. Phase I study of a methanol extract of *Ganoderma lucidum*, edible and medicinal mushroom, in men with mild symptoms of bladder outlet obstruction. Urology 2005;66(suppl. 3A):21.

149. Chen X, Hu ZP, Yang XX, Huang M, Gao Y, Tang W, Chan SY, Dai X, Ye J, Ho PC, Duan W, Yang HY, Zhu YZ and Zhou SF. Monitoring of immune responses to a herbal immuno-modulator in patients with advanced colorectal cancer. Int Immunopharmacol 2006;6:499–508.
150. Ghafa MA, Golliday E, Bingham J, Mansukhani MM, Anastasiasis AG and Katz AE. Regression of prostate cancer following administration of Genistein Combined Polysaccharide (GCP), a nutritional supplement: a case report. J Altern Complement Med 2002;8:493–497.
151. Gao Y, Zhou S, Jiang W, Huang M and Dai X. Effects of ganopoly (a *Ganoderma lucidum* polysaccharide extract) on the immune functions in advanced-stage cancer patients. Immunol Invest 2003;32:201–215.
152. Gao Y, Tang W, Dai X, Chen G, Ye J, Chen E, Koh HL, Li X, and Zhou S, Effects of water-soluble *Ganoderma lucidum* polysaccharides on the immune functions of patients with advanced lung cancer. J Med Food 2005;8:159–168.
153. Chan WK, Lam DT, Law HK, Wong WT, Leung KMW, Lau YL and Chan GC. *Ganoderma lucidum* mycelium and spore extracts as natural adjuvants for immunotherapy. J Altern Complement Med 2005;11:1047–1057.